PHYSICS

1) A long straight wire carries a current of 35 A. What is the magnitude of the field B at a point 20 cm from the wire ?

(1) 3.5×10^{-5} T

(2) 7×10^{-5} T

(3) 10^{-5} T

(4) 2×10^{-5} T

2) The magnetic field B_0 due to current carrying circular loop of radius 12 cm at its centre is 0.50×10^{-4} T. The magnetic field due to this loop at a point on the axis at a distance of 5 cm from the centre is -

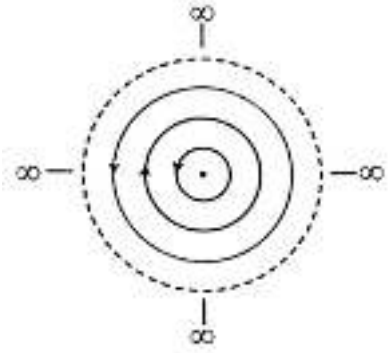
(1) 3.5×10^{-9} T

(2) 5.3×10^{-9} T

(3) 1.3×10^{-5} T

(4) 3.9×10^{-5} T

3) In figure, infinite conducting rings each having current i in the direction shown are placed concentrically in the same plane as shown in the figure. The radius of rings are r, 2r, 2^2r,



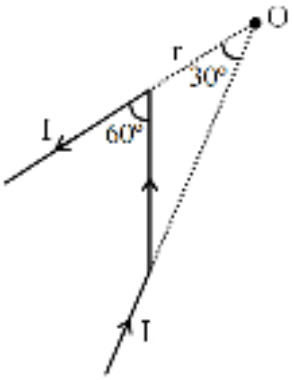
2^3r.....∞. The magnetic field at the centre of rings will be

(1) Zero

(2) $\frac{\mu_0 i}{r}$

(3) $\frac{\mu_0 i}{2r}$

(4) $\frac{\mu_0 i}{3r}$

4) 

Net magnetic field at 'O'.

(1) Zero

(2) $\frac{\mu_0 I}{\sqrt{3}\pi r}$

(3) $\frac{\mu_0 I}{2\sqrt{3}\pi r}\left[\frac{\sqrt{3}-1}{2}\right]$

(4) None

5) A charged particle moves through a magnetic field perpendicular to its direction. Then-

(1) the momentum changes but the kinetic energy is constant

(2) both momentum and kinetic energy of the particle are not constant

(3) both momentum and kinetic energy of the particle are constant

(4) kinetic energy changes but the momentum is constant

6) A closely wound solenoid 80 cm long has 5 layers of winding of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A, the magnitude of $\vec{B}$ inside the solenoid near its centre will be :

(1) $8\pi \times 10^{-3}$ T

(2) $6\pi \times 10^{-3}$ T

(3) $4\pi \times 10^{-3}$ T

(4) $3\pi \times 10^{-3}$ T

7) A toroid of core of inner radius 0.25 m and outer radius 0.26 m around which 3500 turns of a wire are wound. If the current in the wire is 11 A, then magnetic field inside the core of the toroid is

(1) 3×10^{2} T

(2) 3×10^{-2} T

(3) 3×10^{-7} T

(4) 3×10^{7} T

8) Flux ϕ (in weber) in a closed circuit of resistance 10 ohm varies with time t (in *sec*) according to the equation $\phi = 6t^2 - 5t + 1$. What is the magnitude of the induced current at t = 0.25 s ?

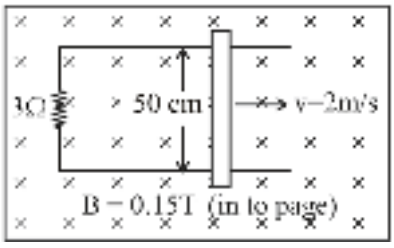
(1) 1.2 A

(2) 0.8 A

(3) 0.6 A

(4) 0.2 A

9) A metallic rod completes its circuit as shown in the figure. The circuit is normal to a magnetic field of B = 0.15 tesla. If the resistances of the rod is negligible then force required to move the rod with a constant velocity of 2 m/sec is:



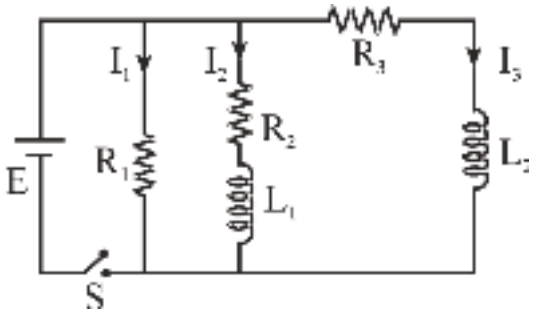
(1) 3.75×10^{-3} N

(2) 3.75×10^{-2} N

(3) 3.75×10^{2} N

(4) 3.75×10^{-4} N

10) Figure shows an L-R circuit. When the switch S is closed, the current through resistor R_1, R_2 and R_3 are I_1, I_2 and I_3 respectively. The value of I_1, I_2 and I_3 at t = 0 s is :



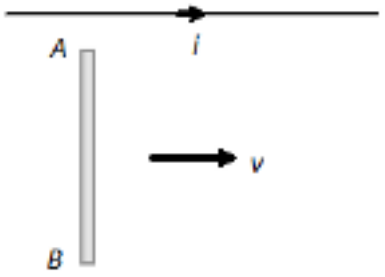
(1) $I_1 = I_2 = I_3 = 0$

(2) $I_1 = \frac{E}{R_1}$, $I_2 = I_3 = 0$

(3) $I_1 = 0$, $I_2 = \frac{E}{R_2}$, $I_3 = \frac{E}{R_3}$

(4) $I_1 = \frac{E}{R_1}$, $I_2 = \frac{E}{R_2 + L_1}$, $I_3 = \frac{E}{R_3 + L_3}$

11) The current carrying wire and the rod AB are in the same plane. The rod moves parallel to the wire with a velocity V. Which one of the following statements is true about induced emf in the rod.



(1) End A will be at lower potential with respect to B

(2) A and B will be at the same potential

(3) There will be no induced e.m.f. in the rod

(4) Potential at A will be higher than that at B

12) A conducting rod of length '□' is moved with a constant velocity v in a magnetic field. A potential difference appears across the two ends

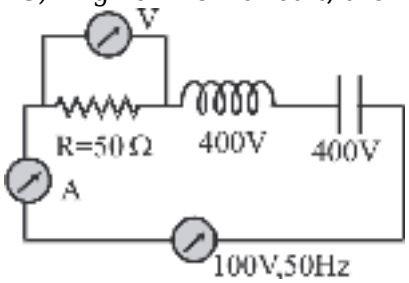
(1) If $\vec{v} \,||\, \vec{\ell}$

(2) If $\vec{v} \,||\, \vec{B}$

(3) If $\vec{\ell} \,||\, \vec{B}$

(4) None of these

13) In given LCR circuit, the voltage across the terminals of a resistance and current will be :–



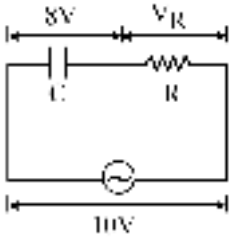
(1) 400V, 2A

(2) 800V, 2A

(3) 100V, 2A

(4) $100\sqrt{2}$V, 2A

14) In a series CR circuit shown in figure, the applied voltage is 10V and the voltage across capacitor is found to be 8V. then the voltage across R, and the phase difference between current and the applied voltage will respectively be



(1) 6V, $\tan^{-1}\left(\frac{4}{3}\right)$

(2) 3V, $\tan^{-1}\left(\frac{3}{4}\right)$

(3) 6V, $\tan^{-1}\left(\frac{5}{3}\right)$

(4) None

15) The charge on a parallel plate capacitor varies as $q = q_0 \cos(2\pi\upsilon t)$. The plates are very large and close together (area = A, distance between plates = d). The displacement current through the capacitor is

(1) $q_0\, 2\pi\upsilon \sin(\pi\upsilon t)$

(2) $-q_0\, 2\pi\upsilon \sin(2\pi\upsilon t)$

(3) $q_0\, 2\pi \sin(\pi\upsilon t)$

(4) $q_0\, \pi\upsilon \sin(2\pi\upsilon t)$

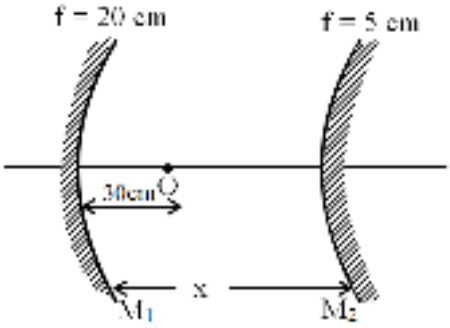
16) If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{j}$T, then what will be expression for electric field?

(1) $\vec{E} = 9\sin(1.6 \times 10^3\, x + 48 \times 10^{10}\, t)\hat{k}\ Vm^{-1}$

(2) $\vec{E} = 60 \sin(1.6 \times 10^3\, x + 48 \times 10^{10}\, t)\hat{k}\ Vm^{-1}$

(3) $\vec{E} = 3 \times 10^{-8} \sin(1.6 \times 10^3\, x + 48 \times 10^{10}\, t)\hat{i}\ Vm^{-1}$

(4) $\vec{E} = 3 \times 10^{-8} \sin(1.6 \times 10^3\, x + 48 \times 10^{10}\, t)\hat{j}\ Vm^{-1}$

17) 

Find out the distance x between both the mirror such that light rays retrace its path after second reflection (consider first reflection on mirror M_1)

(1) 40 cm

(2) 50 cm

(3) 60 cm

(4) 55 cm

18) A prism with refracting angle 60° deviates a light ray by 30°. (When placed in air) The refractive index of prism material is :

(1) $\mu = \sqrt{2}$

(2) $\mu \le \sqrt{2}$

(3) $\mu \ge \sqrt{2}$

(4) $\mu < \sqrt{2}$

19) A short linear object of length b lies along the axis of a concave mirror of focal length f at a distance u from the pole of the mirror. The size of the image is approximately equal to

(1) $b\left(\frac{u-f}{f}\right)^{1/2}$

(2) $_{b}\left(\frac{f}{u-f}\right)^{1/2}$

(3) $_{b}\left(\frac{u-f}{f}\right)$

(4) $_{b}\left(\frac{f}{u-f}\right)^{2}$

20) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A) : Refractive index of glass is higher than that of air.

Reason (R) : Optical density of a medium is directly proportionate to its mass density which results in a proportionate refractive index.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

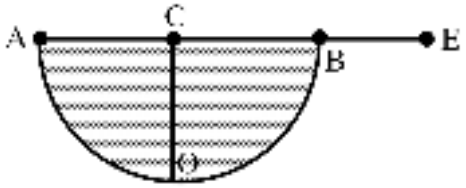
(1) **(A)** is not correct but **(R)** is correct

(2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(3) **(A)** is correct but **(R)** is not correct

(4) Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**

21) A hemispherical vessel is completely filled with a liquid of refractive index μ. A small coin is kept at the lowest point (O) of the vessel as shown in figure. The minimum value of the refractive index of the liquid so that a person can see the coin from point E (at the level of the vessel) is :



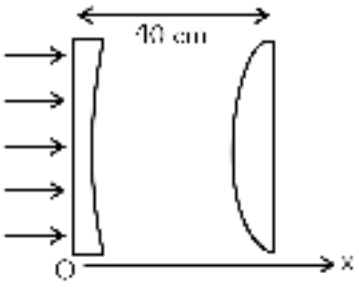
(1) $\sqrt{3}$

(2) $\frac{3}{2}$

(3) $\sqrt{2}$

(4) $\frac{\sqrt{3}}{2}$

22) As shown in the figure, a combination of a thin plano concave lens and a thin plano convex lens is used to image an object placed at infinity. The radius of curvature of both the lenses is 30 cm and refractive index of the material for both the lenses is 1.75. Both the lenses are placed at distance of 40 cm from each other. Due to the combination, the image of the object is formed at distance x cm, from plano concave lens. Then x is :



(1) 40

(2) 80

(3) 100

(4) 120

23) In a Young's double slit experiment the intensity at a point where the path-difference is $\frac{\lambda}{6}$ (λ being the wavelength of the light used) is I. If I^0 denotes the maximum intensity, $\frac{I}{I_0}$ is equal to -

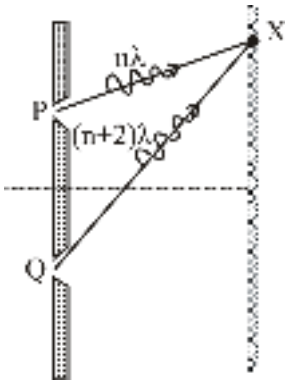
(1) $\frac{1}{\sqrt{2}}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{2}$

(4) $\frac{3}{4}$

24) The figure shows a double slit experiment where P and Q are the slits. The path lengths PX and QX are $n\lambda$ and $(n + 2)\lambda$ respectively, where n is a whole number and λ is the wavelength. Taking the central fringe as origin, what is formed at X :-



(1) First bright fringe

(2) First dark fringe

(3) Second bright fringe

(4) Second dark fringe

25) An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then -

(1) reflected light is completely polarized and the angle of reflection is close to 60°

(2) reflected light is partially polarized and the angle of reflection is close to 30°

(3) both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30°, respectively

(4) transmitted light is completely polarized with angle of refraction close to 30°

26) A light wave is incident normally over a slit of width 24×10^{-5} cm. The angular position of second dark fringe from the central maxima is 30°. What is the wavelength of light :-

(1) 6000 Å

(2) 5000 Å

(3) 3000 Å

(4) 1500 Å

27) The mass defect in a particular nuclear reaction is 0.3 grams. The amount of energy liberated in kilowatt hours is

(1) 1.5×10^6

(2) 2.5×10^6

(3) 3×10^6

(4) 7.5×10^6

28) An electron of energy 11.2 eV undergoes an inelastic collision with a hydrogen atom in its ground state. [Neglet recoiling of atom as $m_H >> m_e$]. Then in this case:

(1) The outgoing electron has energy 11.2 eV

(2) The entire energy is absorbed by the H-atom and the electron stop.

(3) 10.2 eV of the incident electron energy is absorbed by the H-atom and electron would come out with 1.0 eV energy.

(4) None of the above

29) In List-I, we have listed certain quantities for an electron in Bohr atom. Match their dependence on n & Z in list-II.

List-I		**List-II**	
(P)	magnetic dipole moment	(1)	Z^0
(Q)	magnetic field at center of circular path	(2)	Z^2
(R)	angular velocity	(3)	Z^3
(S)	potential energy (assuming standard reference point)	(4)	n^1
		(5)	n^{-3}
		(6)	n^{-2}

(1) P → 1,4;Q → 3,5;R → 2,5;S → 2,6

(2) P → 1,4;Q → 3;R → 2,5;S → 2,6

(3) P → 1,6;Q → 3;R → 2,5;S → 2,6

(4) P → 1,4;Q → 3;R → 3,5;S → 2,5

30) An electron of stationary hydrogen atom passes from the fifth energy level to the first excited level. The velocity that the atom acquired as a result of photon emission will be :

(m is the mass of the atom, R, Rydberg constant and h Planck's constant)

(1) $\frac{24hR}{25m}$

(2) $\frac{35hR}{36m}$

(3) $\frac{21hR}{100m}$

(4) $\frac{2hR}{9m}$

31) The de-Broglie wavelength and kinetic energy of a particle is 2000 Å and 1 eV respectively. If its kinetic energy becomes 1 MeV, then its de-Broglie wavelength becomes :-

(1) 1 Å

(2) 2 Å

(3) 5 Å

(4) 10 Å

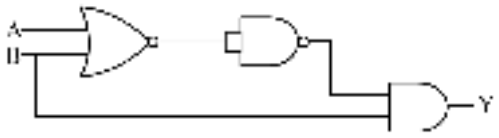
32) ***Assertion (A) :-*** The direction of diffusion current in a junction diode is from P-region to N-region.
Reason (R) :- The majority current carriers diffuse from a region of higher concentration to lower concentration.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

33) The output Y of the given circuit is :- 

(1) A + B

(2) A . B

(3) A

(4) B

34) Zener diode have :

(1) lightly doped and wide depletion layer

(2) heavily doped and narrow depletion layer

(3) lightly doped and narrow depletion layer

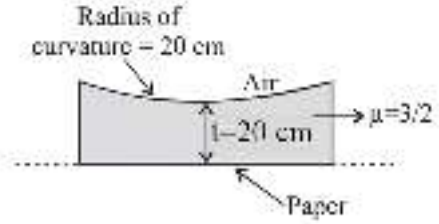
(4) heavily doped and wide deplection layer

35) A series LCR circuit containing 8.0 H inductor, 50 μF capacitor and 80 Ω resistor is connected to 220 V variable frequency ac source. The angular frequencies of the source at which power

transferred to the circuit is half the power at resonant angular frequency are likely to be:

(1) 25 rad/s and 75 rad/s
(2) 30 rad/s and 50 rad/s
(3) 40 rad/s and 60 rad/s
(4) 45 rad/s and 55 rad/s

36) A plano-concave lens is placed on a paper on which a flower is drawn. how far above its actual



position does the flower appear to be ?

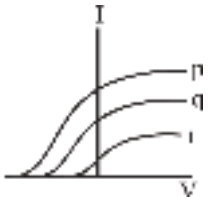
(1) 10 cm
(2) 15 cm
(3) 50
(4) None

37) A photon of energy 8 eV is incident on a metal surface of threshold frequency 1.6×10^{15} Hz. The maximum kinetic energy of photoelectrons emitted is -
(Take h = 6.6×10^{-34} J-s; 1 eV = 1.6×10^{-19} J)

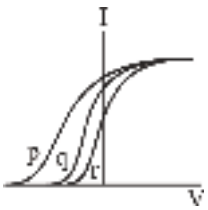
(1) 1.4 eV
(2) 0.8 eV
(3) 4.2 eV
(4) 2.8 eV

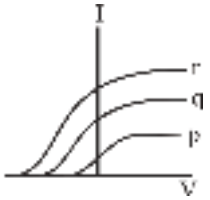
38) Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions ϕ_p = 2.0 eV, ϕ_q = 2.5 eV and ϕ_r = 3.0 eV, respectively. Three light beam of wavelengths 560 nm, 450 nm and 350 nm with equal intensities illuminates the plates of p,q and r respectively. The correct I–V graph for the experiment is :

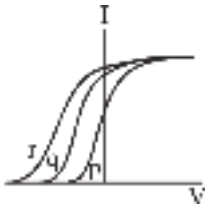
(1)



(2)



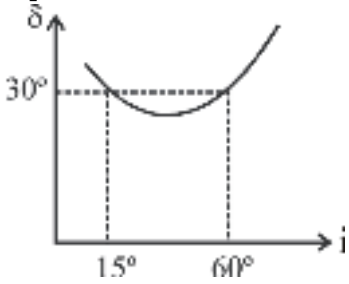
(3) 

(4) 

39) In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 volt battery and a high resistance of 11 kΩ are used. The figure of merit of the galvanometer is 60 μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of θ = 9 divisions when current flows in the circuit. The value of shunt resistance that can cause deflection of θ/2 is closest to :

(1) 55 Ω
(2) 110 Ω
(3) 220 Ω
(4) 550 Ω

40) Figure shows graph of deviation (δ) versus angle of incidence (i) for a light ray striking a prism.

Angle of prism is :- 

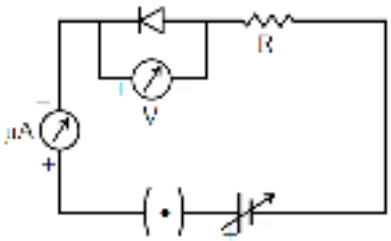
(1) 30°
(2) 45°
(3) 60°
(4) 75°

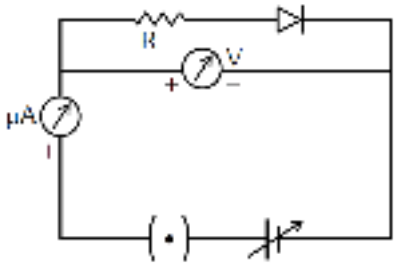
41) A travelling microscope is focussed on a mark on a piece of paper and the scale reading is 5.25 cm. A rectangular block of glass is placed on paper, and the microscope is raised and focused again on the mark, then microscope reading is 11.25 cm. Now a layer of lycopodium powder is sprinkled on the block of glass and viewed the top surface. The reading obtained is 23.25 cm. The refractive index of material of block is :

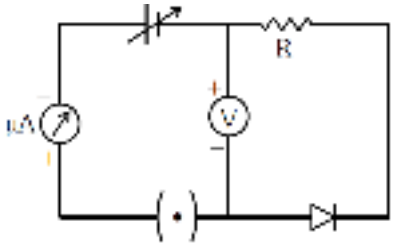
(1) 1.32
(2) 1.45
(3) 1.5
(4) 1.64

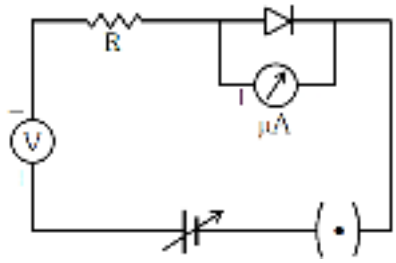
42) Which one is the correct circuit diagram to draw VI curve for reverse bias PN junction diode

characteristic.

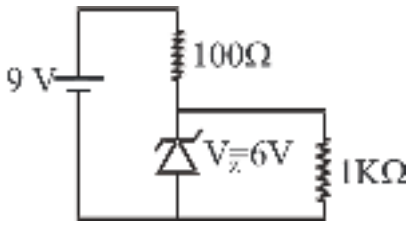
(1) 

(2) 

(3) 

(4) 

43) In the given circuit, current through the zener diode is :-



(1) 30 mA

(2) 24 mA

(3) 36 mA

(4) 6mA

44) Arvind need to identify the electrical components from group of components labelled as A, B, C and D. He performed some simple tests and obtained following observations :
(a) A has silver ring at its one end, with two terminals. It can conduct electricity only in one direction.
(b) B is connected to d.c. source, but current gradually decreases and becomes zero. It is two terminal device and behaves same in both directions.
(c) C is a two terminal device, which conduct electricity only in one direction, with emitting light.
(d) D is two terminal device with several coloured rings on it. It conducts current in both directions

with same steady current.
The correct identification of A, B, C and D are respectively :

(1) Resistor, Photodiode, Diode, Capacitor

(2) Diode, LED, Capacitor, Resistor

(3) Diode, Capacitor, LED, Resistor

(4) LED, Capacitor, Diode, Resistor

45) A Galileo telescope has an objective of focal length 100 cm & magnifying power 50. The distance between the two lenses in normal adjustment will be :

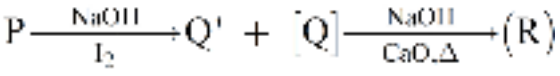
(1) 150 cm

(2) 100 cm

(3) 98 cm

(4) 200 cm

CHEMISTRY

1)  If (P) is first member of Ketone family then choose the correct option ?

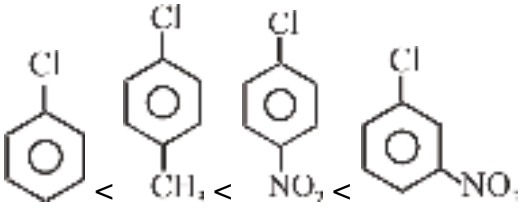
(1) (R) is ethane gas

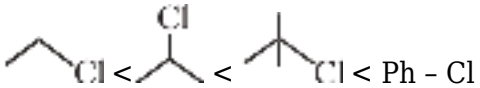
(2) (Q)' is used as anaesthetic

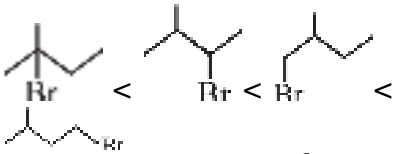
(3) (P) has IUPAC name butanone

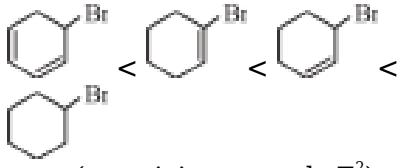
(4) Formation of (R) involves carbanion intermediate

2) Which is correct statement :-

(1) 
(reactivity towards hydrolysis)

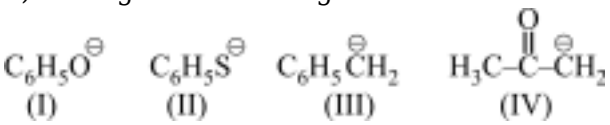
(2)  < Ph – Cl
(reactivity towards SN^1)

(3) 
(reactivity for SN^2)

(4)  < < <

(reactivity towards E^2)

3) Arrange the following in their correct order of nucleophilicity :-



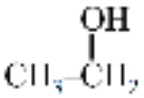
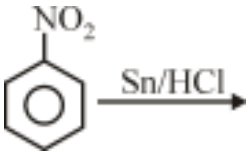
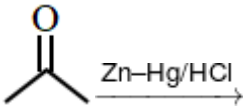
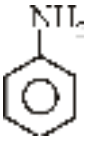
(1) II > III > IV > I

(2) III > IV > I > II

(3) I > IV > II > III

(4) II > IV > III > I

4) Match the column-

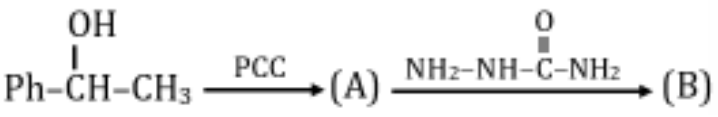
	Column-I		Column-II
(a)	$CH_3-CN \xrightarrow{LiAlH_4}$	(i)	
(b)	$CH_3-COOH \xrightarrow{LiAlH_4}$	(ii)	$CH_3-CH_2-NH_2$
(c)		(iii)	
(d)		(iv)	

(1) (a–i), (b–ii), (c–iii), (d–iv)

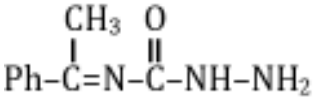
(2) (a–ii), (b–i), (c–iv), (d–iii)

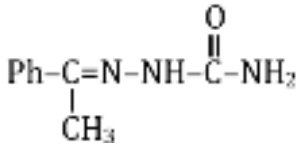
(3) (a–ii), (b–i), (c–iii), (d–iv)

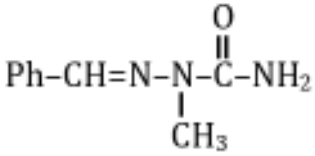
(4) (a–i), (b–iii), (c–ii), (d–iv)

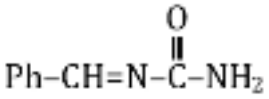
5) 

Product (B) is :-

(1) 

(2) 

(3) 

(4) 

6) Match the column with reagent and their product.

	Column-I		**Column-II**
A	NH_2–NH_2	p	Schiff base
B	$NH_2NHCONH_2$	q	Hydrazone
C	2° amine	r	Enamine
D	1° amine	s	Semicarbazone

(1) A–q; B–p; C–r; D–s

(2) A–q; B–s; C–p; D–r

(3) A–q; B–s; C–r D–p

(4) A–q; B–r; C–p; D–s

7) Match **List-I** with **List-II**

List–I (Reagent)		**List-II (Name of the reaction)**	
A.	H_2, Pd-$BaSO_4$	I.	Gattermann-Koch reaction
B.	(i) CrO_2Cl_2/CS_2; (ii) H_2O	II.	Reimer-Tiemann reaction
C.	CO, HCl, Anhyd. $AlCl_3$/CuCl	III.	Etard reaction
D.	$CHCl_3$,NaOH	IV.	Rosenmund reduction

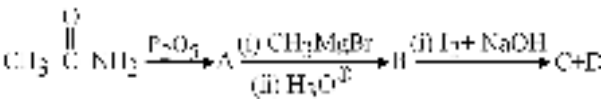
Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

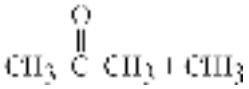
(2) A-I, B-III, C-IV, D-II

(3) A-II, B-I, C-III, D-IV

(4) A-II, B-III, C-I, D-IV

8) 

Product C and D respectively are (where D is yellow ppt) :-

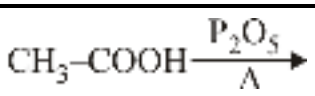
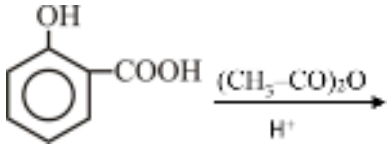
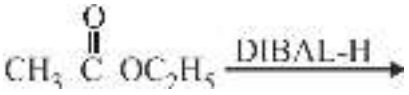
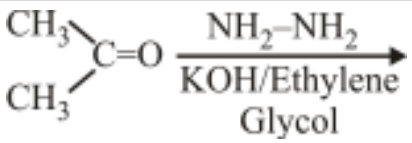
(1) 

(2) CH_3–COOH + CHI_3

(3) CHI_3 + CH_3–$COO^{\ominus}$

(4) CH_3–$COO^{\ominus}$+CHI_3

9) Match the column :-

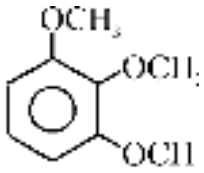
Column-I (Reaction)		Column-II (Product)	
A.		P.	Analgesic
B.		Q.	Anhydride
C.		R.	Alkane
D.		S.	Aldehyde

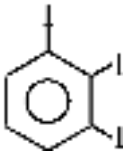
(1) A-Q, B-P, C-S, D-R

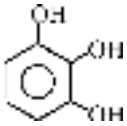
(2) A-P, B-Q, C-S, D-R

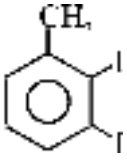
(3) A-Q, B-P, C-R, D-S

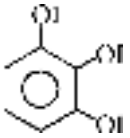
(4) A-Q, B-S, C-P, D-R

10)  Product is :

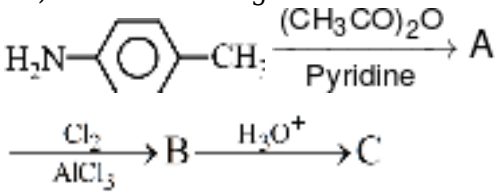
(1) 

(2) 

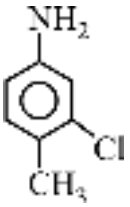
(3) 

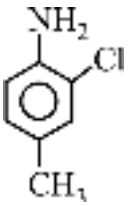
(4) 

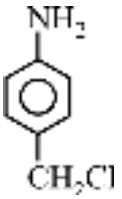
11) In the following reaction

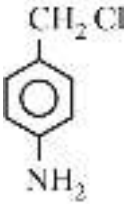


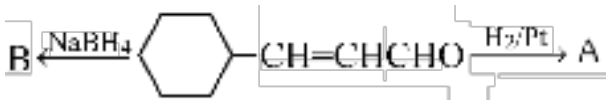
The product 'C' will be

(1) 

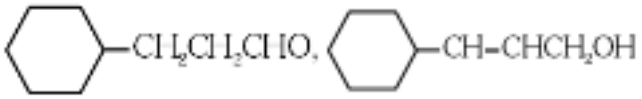
(2) 

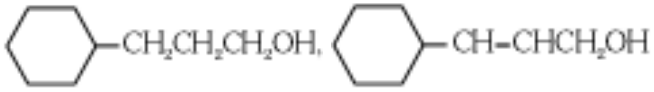
(3) 

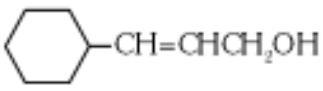
(4) 

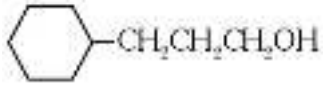
12) 

A and B are respectively :

(1) 

(2) 

(3)  , in both cases

(4)  , in both cases

13) **Assertion :** Polysaccharide are called non-reducing sugars.
Reason : Carbohydrates which yield a large number of monosaccharide units on hydrolysis are called polysaccharide.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

14)

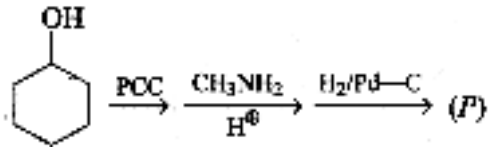
Which of the following is/are correct statement/s?

(1) Nucleotides are joined together by phosphodiester linkage between 5' and 3' carbon atoms of the pentose sugar.

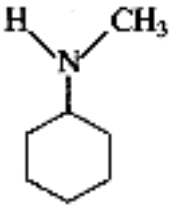
(2) The two strands in DNA are complementary to each other because the hydrogen bonds are formed between specific pairs of bases.

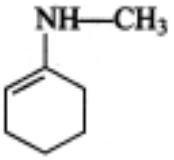
(3) DNA molecule is capable of self duplication during cell division and identical DNA strands are transferred to daughter cells.

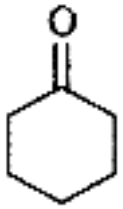
(4) All of these

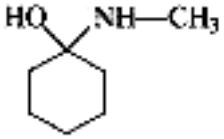
15) 

P will be :

(1) 

(2) 

(3) 

(4) 

16) Which of the following combination is suitable for the preparation of 3-methyl-2-butanol ?

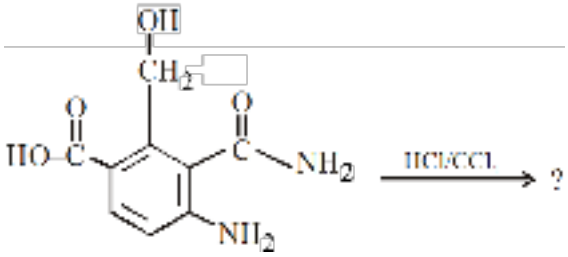
(1) 2-Butanone+methyl magnesium bromide

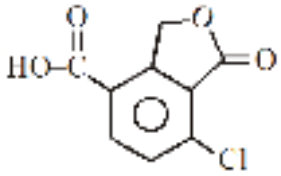
(2) Acetone+ethyl magnesium bromide

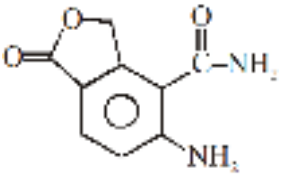
(3) Acetaldehyde + isopropyl magnesium bromide

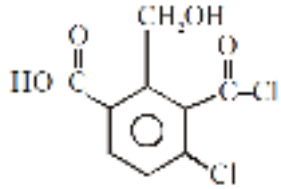
(4) Ethyl propionate + methyl magnesium bromide

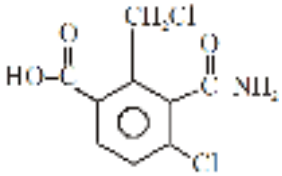
17) The main product of the following reaction will be



(1) 

(2) 

(3) 

(4) 

18) **Statement-I** : Formaldehyde give both aldol & cannizaro reaction.
Statement-II : Benzaldehyde give haloform reaction.

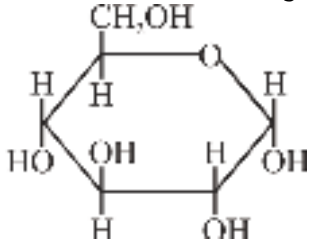
(1) Both **Statement I** and **Statement II** are incorrect

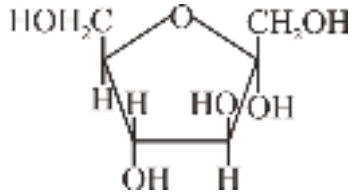
(2) **Statement I** is correct but **Statement II** is incorrect

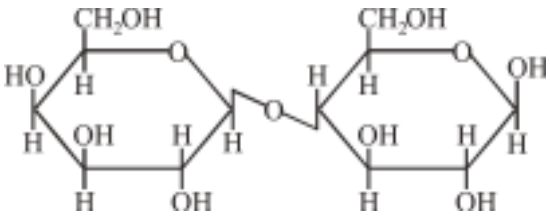
(3) Both **Statement I** and **Statement II** are correct

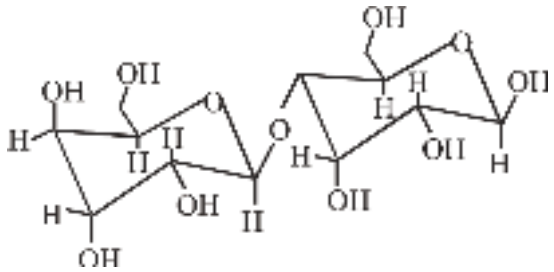
(4) **Statement I** is incorrect but **Statement II** is correct

19) Which of the following is a reducing sugar :-

(a) 

(b) 

(c) 

(d) 

(1) a, b, d

(2) b, c, d

(3) b, c

(4) All of these

20) Consider following table–

Column-I

(A) Best and latest technique for isolation, purification and separation of organic compounds

(B) In lassaigne's test for 'N'

(C) In Lassaigne's test for 'S'

(D) Aniline is purified from water by

(E) Glycerol is purified from spent lye

Column-II

(i) Chromatography

(ii) Vacuum distillation

(iii) Violet coloured complex is formed

(iv) Prussian blue coloured complex is formed

(v) Steam disstillation

The correct match is–

	(A)	(B)	(C)	(D)	(E)
(1)	(i)	(ii)	(iii)	(iv)	(v)
(2)	(iv)	(v)	(iii)	(ii)	(i)
(3)	(i)	(iv)	(iii)	(v)	(ii)
(4)	(v)	(iii)	(iv)	(ii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

21) In the kjeldahl's method For estimation of nitrogen present in a soil sample, ammonia evolved From 0.75 g of sample neutralized 10 ml of 1 M H_2SO_4. The percentage of nitrogen in the soil is :-

(1) 37.33
(2) 45.33
(3) 35.33
(4) 43.33

22) At 0°C temperature reaction of aniline with HNO_2 and subsequent treatment with alkaline β–naphthol solution produces a precipitate whose colour would be :-

(1) Black
(2) Purple
(3) White
(4) Orange

23) Which of the following are peroxo acids of sulphur:-

(1) H_2SO_5 & $H_2S_2O_8$
(2) H_2SO_5 & $H_2S_2O_7$
(3) $H_2S_2O_7$ & $H_2S_2O_8$
(4) $H_2S_2O_6$ & H_2SO_4

24) Which of the following metal react with NaOH to produce H_2 :-

(1) Mg
(2) Ba
(3) Zn
(4) Cu

25) The group reagent for the detection of basic radicals in IVth group.

(1) $NH_4OH + (NH_4)_2CO_3$
(2) $NH_4OH + Na_2HPO_4$
(3) $NH_4OH + H_2S$ gas
(4) dil HCl + H_2S gas

26) Which of the following can be detected by borax bead test?

(1) Cu^{2+}
(2) Zn^{2+}
(3) Pb^{2+}
(4) None of these

27) An ion gives red precipitate on addition of DMG ion is.

(1) Cu^{+2}

(2) Fe^{+2}

(3) Ni^{+2}

(4) Co^{+2}

28) Which do not give chromyl chloride test.

(1) $CuCl_2$

(2) $HgCl_2$

(3) $ZnCl_2$

(4) $C_6H_5NH_3^+Cl^-$

29) Blue lake test is given by -

(1) Fe^{3+}

(2) Al^{3+}

(3) $CuSO_4.5H_2O$

(4) Cr^{3+}

30) When salt 'A' reacts with acidified $K_2Cr_2O_7$ it gives orange red gas 'B' which is passed to NaOH solution and gives yellow soluble compound 'C'. Yellow ppt 'D' is obtained on adding lead acetate solution to 'C' . 'A' gives brick red colour in flame test. A, B, C and D are respectively.

(1) $CaSO_4$, CrO_2Cl_2, $Na_2Cr_2O_7$, $PbCr_2O_7$

(2) $BaCl_2$, CrO_2Cl_2, Na_2CrO_4, $PbCrO_4$

(3) $CaCl_2$, CrO_2Cl_2, Na_2CrO_4, $PbCrO_4$

(4) $BaSO_4$, CrO_2Cl_2, $CaCrO_4$, $PbCrO_4$

31) Which set of anions gives same colour gas with dilute acid and concentrated acid respectively.

(1) Br^-, NO_3^-

(2) NO_2^- , NO_3^-

(3) I^-, NO_2^-

(4) NO_2^- , Br^-

32) $K_4[Fe(CN)_6]$ can be used to detect one or more out of Fe^{+2}, Fe^{+3}, Zn^{+2}, Cu^{+2}

(1) Only Fe^{+2}, Fe^{+3}

(2) Only Fe^{+3}, Zn^{+2}, Cu^{+2}

(3) All but not Cu^{+2}

(4) All of these

33) Which halide does **NOT** produce halogen when heated with MnO_2 in dil. H_2SO_4 :-

(1) F^-

(2) Cl^-

(3) Br^-

(4) I^-

34) Atomic number of Ag is 47. In the same group, the atomic numbers of elements placed above and below Ag will be:

(1) 37, 67

(2) 29,79

(3) 39, 69

(4) 29, 65

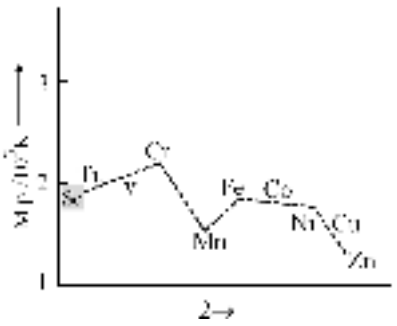
35) In which of the following reaction hydridisation of central atom is not change

(1) $PCl_5 + 5H_2O \xrightarrow{\Delta}$

(2) $SF_4 + 4H_2O \xrightarrow{\Delta}$

(3) $XeF_6 + 6H_2O \xrightarrow{\Delta}$

(4) $SiCl_4 + 4H_2O \xrightarrow{\Delta}$

36) Consider the following graph 

The melting point of Mn is exceptionally very low due to

(1) Very high IE of Mn

(2) Stable electronic configuration of Mn

(3) Less delocalisation of e^- in Mn

(4) Very weak metallic bond in Mn

37) Match the properties given in Column I with the metals given in Column II -

	Column-I (Property)		**Column-II (Metal)**
(i)	An element which x can show +8 oxidation state	(a)	Mn
(ii)	3d block element that can show upto +7 oxidation state	(b)	Cr

(iii)	3d block element with highest melting point	(c)	Os
		(d)	Fe

(1) i-c, ii-a, iii-b

(2) i-a, ii-b, iii-c

(3) i-a, ii-c, iii-d

(4) i-c, ii-b, iii-a

38) Match the following for d-block compound

(a)	Catalytic property	(i)	Multiple oxidation state
(b)	Interstitial compound	(ii)	Neither ionic nor covalent
(c)	Alloy formation	(iii)	Similar radii

(1) a-(i), b-(ii), c-(iii)

(2) a-(ii), b-(iii), c-(i)

(3) a-(iii), b-(i), c-(ii)

(4) None of these

39) Which of the following pair contain all diamagnetic ion ?

(1) Yb^{+2} and Lu^{+3}

(2) Ce^{+4} and Pr^{+4}

(3) Dy^{+3} and Lu^{+3}

(4) Lu^{+3} and Gd^{+3}

40) In the 3d series the enthalpy of atomization of zinc is the lowest (126 kJ mol^{-1}) because?

(1) It have stronger interatomic at interaction

(2) It have high melting point

(3) It have high degree of ionization

(4) In the formation of metallic bonds, no electrons from 3d orbitals are involved in case of zinc

41) Which of the following statements is incorrect?

(1) The order of splitting energy is $PtCl_4^{2-} > PdCl_4^{2-} > NiCl_4^{2-}$

(2) $[Zn(NH_3)_4]^{+2}$ is colourless whereas $\left[Ni(H_2O)_6\right]^{2+}$ is coloured.

(3) $[M(en)(gly)]^{n+}$ will represent geometrical isomerism.

(4) The magnetic moment of $K_3\left[Fe(CN)_6\right]$ is $\sqrt{3}$ B.M.

42) Which is inner orbital and low spin complex ?

(1) $[NiCl_4]^{-2}$

(2) $[Mn(CN)_6]^{-3}$

(3) $[CoF_6]^{-3}$

(4) $[FeF_6]^{-3}$

43) Value of CFSE, in tetrahedral complex having $3d^4$ configuration of metal ion, surrounded by weak field ligands, will be :-

(1) $-\frac{2}{5}\Delta_t$

(2) $+\frac{2}{5}\Delta_t$

(3) $-\frac{4}{5}\Delta_t$

(4) $+\frac{3}{5}\Delta_t$

44) Select bidentate or didentate ligand from the following.

(1) CO

(2) SCN^-

(3) CH_3COO^-

(4) $C_2O_4^{2-}$

45) The IUPAC name of the complex $[Ag(H_2O)_2][Ag(CN)_2]$ is :

(1) diaquasilver(II) dicyanidoargentate(II)

(2) dicyanidosilver(I) diaquaargentate(I)

(3) diaquasilver(I) dicyanidoargentate(I)

(4) dicyanidosilver(II) diaquaargentate(II)

BIOLOGY

1) Which one of the following can be explained on the basis of Mendel's Law of segregation ?
(A) Out of one pair of factors one is dominant and the other is recessive.
(B) Alleles do not show any blending.
(C) Factors occur in pairs in normal diploid plants.
(D) The discrete unit controlling a particular character is called factor.
(E) Both the characters are recovered as such in the F_2 generation though one of these is not seen at the F_1 stage.
Choose the correct answer from the options given below :-

(1) A, B and C only

(2) B and E only

(3) A, C, D and E only

(4) A, B, C, D and E

2) Given below are two statements :
Statement-I: The sex determining mechanism in case of humans is XY type. Out of 23 pairs of chromosomes present, 22 pairs are exactly same in both males and females; these are the autosomes.
Statement-II: A pair of X-chromosomes are present in the female, whereas in male the presence of X and Y chromosome are determinant of the male characteristic. During spermatogenesis among males, two types of gametes are produced.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement I** is incorrect but **Statement II** is correct.

3) A child with blood group genotype I^AI^B is born of a women with genotype I^BI^B. The father could not be a man of genotype.

(1) I^AI^A
(2) I^BI^B
(3) I^AI^B
(4) I^Ai

4) Experimental verification of chromosomal theory of inheritance was given by :-

(1) Sutton and Boveri
(2) Alfred Sturtevant
(3) Hugo devries
(4) Morgan and his colleagues

5) Which of the following genotype pair given below do not show same expression w.r.t. seed shape and seed colour-

(1) RrYy ; RRYy
(2) RRYY ; RrYY
(3) rrYy : rrYY
(4) rrYy ; Rryy

6) A trait controlled by genes which are present at several loci is an example of:

(1) Multiple allelism
(2) Incomplete dominance
(3) Pleiotropy
(4) Polygenic inheritance

7) Consider the following statements for some disease :
(a) Cystic fibrosis is non-mendelian disorder.

(b) Phenylketonuria is a Mendelian disorder.
(c) Colour blindness occurs in 8% females.
(d) Haemophilia is a sex linked Non-Mendelian disorder.
(e) In sickle cell anaemia, mutant haemoglobin molecule undergoes polymerisation under low oxygen tension.
Choose the option with correct statements from given below options :

(1) (b) and (e) only
(2) (b) and (a) only
(3) (e) and (d) only
(4) (c) and (d) only

8) Match the column-I and II :

Column-I		**Column-II**	
I	Down syndrome	A.	Sterile female
II	Klinefelter syndrome	B.	Trisomy of 21
III	Turner syndrome	C.	Gynaecomastia
IV	Aneuploidy	D.	Gain or loss of a chromosome

(1) I–A, II–B, III–C, IV–D
(2) I–B, II–C, III–A, IV–D
(3) I–D, II–C, III–B, IV–A
(4) I–C, II–D, III–B, IV–A

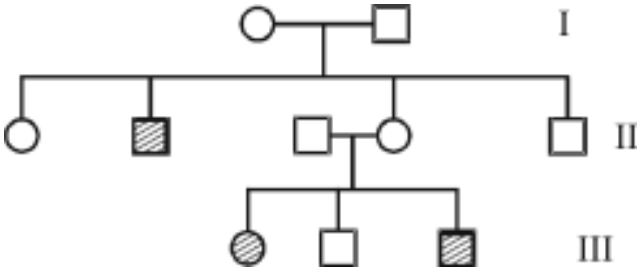
9) **Assertion :** When two genes in a dihybrid cross were situated on the same chromosome, the proportion of parental combination were much higher than the non-parental type.
Reason : This is due to physical association of genes i.e. linkage.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

10) **Statement-I :** Law of segregation is based on the fact that the alleles do not show any blending and both the characters are recovered as such in F_1 generation.
Statement-II : Law of dominance is used to explain the expression of only one of parental character in a monohybrid cross in the F_1 and expression of both in the F_2.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct and Statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

11) Given below is the pedigree of Sickle cell anaemia in a family :-



In this the genotype of both the parents of generation I will be :-

(1) AA & Aa

(2) AA & aa

(3) Aa and Aa

(4) Aa and aa

12) When a phosphate group is attached to __A__ of nucleoside through __B__ bond, a __C__ is formed.

(1) A=3'OH, B- Amide, C=nucleotide

(2) A=3'OH, B-Phosphoester, C=nucleotide

(3) A=5'OH, B- Phosphoester, C=nucleotide

(4) A=5'OH, B-Phosphoamide, C=nucleotide

13) If A and C make 30% and 20% of DNA, respectively, what will be the percentage composition of T and G ?

(1) T : 20%, G : 30%

(2) T : 30%, G : 20%

(3) T : 30%, G : 30%

(4) T : 20%, G : 20%

14) The unequivocal proof that DNA is the genetic material came from the experiment that utilized :

(1) *Streptococcus*

(2) T_2 bacteriophage, *E.coli*

(3) *E.coli*, N^{15}

(4) P^{32} , S^{35}, S and R type of bacteria

15) What is true for Okazaki-segment ?

(1) Polymerize in 5'-3' direction and formed in leading strand

(2) Polymerize in 3'-5' direction and formed in lagging strand

(3) Formed at the time of transcription

(4) Polymerize in 5'-3' direction and formed in lagging strand

16) "The transcription and translation can be coupled in bacteria"
what is the reason behind it ?

(1) In bacteria mRNA does not require any processing

(2) No separation of cytosol and nucleus in bacteria

(3) There are three type of RNA polymerase present in bacteria

(4) Both (1) and (2)

17) **Lac operon** consists :
(a) A monocistronic structural gene which is regulated by a common promoter.
(b) Glucose and galactose can not act as an inducer
(c) The 'y' gene codes for permease which increases permeability of cell to β-galactosides
(d) Allolactose acts as an end product
(e) Proposed by a geneticist F. Jacob and biochemist J. Monod.
How many are incorrect ?

(1) only 'a'

(2) a and d

(3) a, b, d

(4) a, c

18) Match the following :

Column-I		Column-II	
a	George Gamow (a, physicist)	i	*In vitro* translation of homopolymer.
b	H.G. Khorana	ii	Templates independent RNA polymerization (polynucleotide phosphorylase)
c	Nirenberg	iii	Genetic code is triplet (3 bases collectively code for Aminoacid)
d	Severo Ochoa	iv	Chemical method of synthesis of homopolymer and co-polymer.

(1) a(iii), b(iv), c(i), d(ii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(iii), b(i), c(iv), d(ii)

19) **Assertion (A) :** The presence of introns is reminiscent of antiquity and the process of splicing represents the dominance of RNA world.
Reason (R) : The cellular factory responsible for synthesising proteins is ribosome.The ribosome consists of structural RNAs and about 80 different proteins.

(1) Assertion is ture but Reason is false

(2) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.

20) Given below are two statements :-
Statement-I :- In semiconservative DNA replication the two strands would separate and only one strand act as a template for synthesis of new complementary strand
Statement-II :- After the completion of replication, each DNA molecules would have one parental and one newly synthesised strand.
Choose the correct answer from the options given below :-

(1) Statement I and II both are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement II is correct but statement I is incorrect

(4) Statement I and II both are incorrect

21) (I) Translocation of ribosome
(II) Termination of translation
(III) Charging of t-RNA
(IV) m-RNA binds to Ribosome
(V) First peptide bond formation
What is correct sequence of translation process ?

(1) I, II, III, IV, V

(2) III, IV, V, I, II

(3) III, IV, I, V, II

(4) III, IV, V, II, I

22) Which of the following statement is not correct ?

(1) RNA polymerase with the help of sigma factor binds to the promoter and initiate transcription.

(2) Amino acids have structural specialities to read the genetic code.

(3) If two charged tRNAs are brought close enough, the formation of peptide bond takes place between amino acids.

(4) Polynucleotide phosphorylase is helpful in RNA synthesis in template-independent manner.

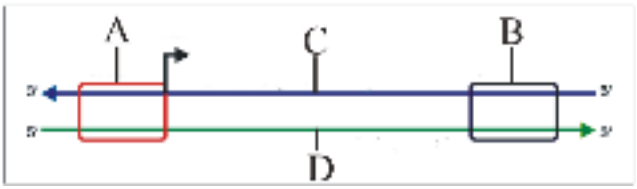
23) Read the given following statements ;
(a) tRNA has an amino acid acceptor end to which it binds to amino acids
(b) tRNA are specific for each amino acid
(c) For initiation, there is another specific tRNA that is referred to as initiator tRNA
(d) There are no tRNAs for stop condons
(e) In actual structure, the tRNA is a compact molecule which looks like a clover leaf
Choose the correct answer from the options given below :-

(1) (b), (d). (e) only

(2) (a), (b). (c), (d) only

(3) (a), (c). (d), (e) only

(4) (a), (c) only

24) Given diagram represent the component of a transcription unit. Select the correct answer regarding it 

	A	B	C	D
(1)	Terminator	Promoter	Template strand	Coding strand
(2)	Promoter	Terminator	Coding strand	Template strand
(3)	Promoter	Terminator	Template strand	Coding strand
(4)	Terminator	Promoter	Non Coding strand	Template strand

(1) 1

(2) 2

(3) 3

(4) 4

25) I = Histone + DNA
II = Solanoid
III = Chromosome,
IV = Nucleosome
V = Chromatin.
Which of the following sequence is correctly indicating the increasing level of arrangement?

(1) I → IV → V → II → III

(2) I → IV → II → V → III

(3) IV → I → V → II → III

(4) IV → I → III → V → II

26) Match the column-I and column-II (about population interactions) and select the correct option-

	Column-I		Column-II
I.	Yucca	(a)	Fungi
II.	Roots of higher plants	(b)	Sea anemone
III.	*Ophrys*	(c)	Tick bird
IV.	Clown fish	(d)	Psuedo copulation

V.	Rhinoceros	(e)	Pronuba moth

(1) I - (e), II - (a), III - (d), IV - (c), V - (b)

(2) I - (e), II - (d), III - (a), IV - (b), V - (c)

(3) I - (a), II - (d), III - (c), IV - (e), V - (b)

(4) I - (e), II - (a), III - (d), IV - (b), V - (c)

27) Given below two statements one is labelled as Assertion (A) and other is labelled as Reason (R) :
Assertion (A) : Gause's competitive exclusion principle states that two closely related species competing for same resources can co-exist indefinitely.
Reason (R) : By resources partitioning two competing species live together by avoiding competition. They reduces competition by choosing different times for feeding or different foraging pattern, this was shown by Macarthur in observation on five closely related species of warbler living on same tree.
In the light of above statement select the correct options :-

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

28) The carrying capacity of a population is determined by its :

(1) Limiting resources

(2) Natality

(3) Mortality

(4) Population Growth Rate

29) Ecology explains us :-

(1) How organisms, while remaining as an individual interact with other organisms.

(2) How organism, interact with physical environment.

(3) As a group how organisms behave like organised population, community, ecosystem or even as the whole biosphere.

(4) All of these

30) In an Area having 200 *Parthenium* plants and a single huge Banyan tree, population density is measured in terms of :-

(1) Biomass only

(2) Percent cover only

(3) Biomass or percent cover

(4) Number, Biomass or percent cover

31)

Which of the following is correct for r–selected species?

(1) Large number of progeny with small size

(2) Large number of progeny with large size

(3) Small number of progeny with small size

(4) Small number of progeny with large size

32) Read the following three statements (A-C).
(A) Geometric growth produces J-shaped population growth curve.
(B) Logistic growth occurs when resources are limiting.
(C) Equation for exponential growth is $N_t = N_o e^{rt}$
How many of the above statements are true ?

(1) Zero

(2) One

(3) Two

(4) Three

33) Match List-I with List-II

	List-I (Interactions)		**List-II (Species A and Species B)**
(A)	Parasitism	(I)	+(A), +(B)
(B)	Mutualism	(II)	+(A), 0(B)
(C)	Commensalism	(III)	–(A), –(B)
(D)	Competition	(IV)	+(A), –(B)

Choose the correct answer from the options given below :

(1) A-IV, B-I, C-II, D-III

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-II, C-I, D-III

34) Given below are two statements:
Statement-I : Connel's elegant field experiments showed that on the rocky sea coast of Scotland, the larger and competitively superior barnacle Balanus dominantes the intertidal area and excludes the smaller barnacle chathalamus from that zone.
Statement-II : The life cycle of parasites is often simple, involving one or two intermediate host or vectors to facilitate parasitisation of its primary host.
In the light of above statements, choose the correct answers from the options given below :

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

35) Mr. X is eating curd. For this he is occupying trophic level :-

(1) First
(2) Second
(3) Third
(4) Fourth

36) The ecological pyramid of number in pond ecosystem is -

(1) Upright
(2) Inverted
(3) May upright or Inverted
(4) First upright then inverted

37) Limitation of ecological pyramids is/are :-

(1) It does not take into account the same species belonging to two or more trophic levels.
(2) It assumes a simple food chain, it does not accommodate a food web.
(3) Saprophytes are not given any place.
(4) All of the above.

38) How many statement are incorrect :-
(a) The number of trophic levels in the grazing food chain is restricted.
(b) Organism at each trophic level depend on those at the higher trophic level for their energy demands
(c) Each trophic level has a certain mass of living material at a particular time called as the standing crop
(d) Decomposers has the largest population in grazing food chain

(1) Three
(2) Four
(3) Two
(4) One

39) Match the Column I with Column II :-

	Column I		**Column II**
(A)	Accumulation of dark coloured amorphous substance	(i)	Fragmentation
(B)	Water soluble nutrients go down into the soil horizon	(ii)	Humification
(C)	Release of inorganic nutrients by microbial activity	(iii)	Mineralisation
(D)	Break down of detritus into smaller particles	(iv)	Leaching

	A	B	C	D
(1)	(ii)	(iii)	(iv)	(i)
(2)	(ii)	(iv)	(iii)	(i)
(3)	(i)	(iii)	(iv)	(ii)
(4)	(iv)	(i)	(ii)	(iii)

(1) 1

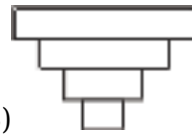
(2) 2

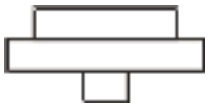
(3) 3

(4) 4

40) Which of the following representation shows the pyramid of biomass in lake ecosystem?

(A) 

(B) 

(C) 

(D) 

(1) A

(2) B

(3) C

(4) D

41) Which of the following is an in-situ conservation method ?

(1) National park

(2) Tissue culture

(3) Zoological park

(4) Botanical garden

42) Choose the correct match for the following :-

	Column-I		Column-II
a.	Jhoom cultivation	i.	8.1% of the world
b.	Sacred grove	ii.	In-situ conservation
c.	Land area of India	iii.	2.4% of the world
d.	Biodiversity of India	iv.	Deforestation

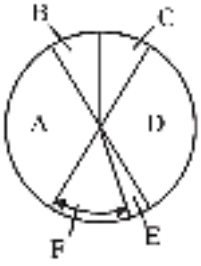
(1) a-iv, b-ii, c-i, d-iii

(2) a-ii, b-iv, c-iii, d-i

(3) a-ii, b-iv, c-i, d-iii

(4) a-iv, b-ii, c-iii, d-i

43) Study the following pie chart with respect to global diversity of plants and select the correct option :-



	A	B	C	D	E	F
(1)	Angiosperm	Lichen	Algae	Fungi	Mosses	Fern & Allies
(2)	Fungi	Lichen	Algae	Angiosperm	Mosses	Fern & Allies
(3)	Fungi	Mosses	Fern & Allies	Angiosperm	Lichen	Algae
(4)	Angiosperm	Fern & Allies	Lichen	Fungi	Algae	Mosses

(1) 1

(2) 2

(3) 3

(4) 4

44) Species diversity increase from :

(1) Low altitude to high altitude and from low latitude to high latitude.

(2) High altitude to low altitude, and from high latitude to low latitude.

(3) High altitude to low altitude, and from low latitude to high latitude.

(4) Low altitude to high altitude, and from high latitude to low latitude.

45)

Which of the following statement(s) is/are correct :-
(A) Among Animals, insects are the most species rich taxonomic group.
(B) Conventional taxonomic methods are suitable for Identification of microbial species.
(C) Speciation is generally a function of time.
(D) The last twenty years alone have witnessed the disappearance of 27 species.

(1) A, C and D

(2) A, B and C

(3) A, B, C and D

(4) B, C and D

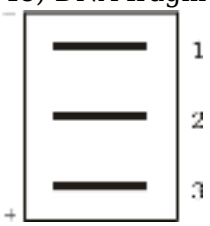
46) Which of the following method can be termed as vectorless approach to introduce foreign DNA into host cells :-
(A) Microinjection
(B) Biolistics
(C) Bacterial Artifical chromosome

(1) A only
(2) A & C both
(3) C only
(4) A & B both

47) *Agrobacterium tumefaciens* is known as :-

(1) Natural genetic engineer of plant
(2) Natural genetic engineer of animal
(3) Symbiotic bacteria
(4) Symbiotic plant

48) DNA fragments separated by gel electrophoresis are shown. Mark the correct statement :-



(1) Band '3' contains more positively charged DNA molecule than '1'
(2) Band '3' has longer DNA fragment than '1' and '2'
(3) Band '1' has longer DNA fragment than '2' and '3'
(4) All the bands have equal length and charges but differ in base composition

49) Choose the correct matches :-

(a)	First Restriction Endonuclease	Hind II
(b)	Eco RI and Eco RV	Sticky ends producing
(c)	cDNA	Shorter than original gene
(d)	Direct method of gene transfer	*Agrobacterium* mediated.
(e)	Conventional methods of diagnosis	Serum and urine analysis
(f)	PCR	RNA primers

(1) a, c, f

(2) b, d, f

(3) a, c, e

(4) b, c, e

50) Which one is incorrect with respect to gel electrophoresis :-

(1) Wells for DNA sample loading are made at anode

(2) Largest fragments of DNA lie far away from cathode

(3) Smallest fragments of DNA lie near to anode

(4) Both 1 and 2

51) **Assertion :-** In gel electrophoresis, the most commonly used matrix is agarose.
Reason :- The DNA fragments separate (resolve) according to their size through sieving effect provided by the agarose gel.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

52)

Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence.
A. Insertion of recombinant DNA into the host cell.
B. Cutting of DNA at specific location by restriction enzyme.
C. Isolation of desired DNA fragment.
D. Amplification of gene of interest using PCR.

(1) C, A, B, D

(2) C, B, D, A

(3) B, D, A, C

(4) B, C, D, A

53) Find mismatch :-

(1) Cry I Ac – Boll worms

(2) Cry II Ab – Boll worms

(3) Cry II Ab – Corn borer

(4) Cry I Ab – Corn borer

54) First transgenic cow Roise milk is rich in which protein:

(1) Human alpha globin

(2) Human alpha lactalbumin

(3) Human beta lactalbumin

(4) Human beta globin chain

55) Silencing of mRNA has been used in producing transgenic plants resistant to :

(1) Bacterial blights

(2) Bollworms

(3) Nematodes

(4) White rusts

56) Fill up the blanks.
I. Using *Agrobacterium* vectors, ...A.... specific genes are introduced into the host plant.
II. RNAi takes place in all eukaryotic organisms as a method of .....B.....
III. The genes .....C.... and .....D.... control cotton bollworms.
IV. ....E... has been introduced in Bt corn to protect the same from corn borer.
A to E in above statements can be:-

(1) A-fungal, B-pest resistance,
C-cry II Ab, D-cry II Ab, E-cry I Ac

(2) A-nematode, B- Cellular defence
C-cry I Ac, D-cry II Ab, E-cry I Ab

(3) A-insects, B-cellular defense,
C-cry II Ab, D-cry I Ac, E-cry I Ab

(4) A-virus, B-protection against viral infection,
C-cry I Ac, D-cry II Ab, E-cry II Ac

57)

Match List I with List II :

List I		**List II**	
A.	α-1 antitrypsin	I.	Cotton bollworm
B.	Cry IAb	II.	ADA deficiency
C.	Cry IAc	III.	Emphysema
D.	Enzyme replacement therapy	IV.	Corn borer

Choose the correct answer from the options given below:

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-I, C-II, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-IV, C-I, D-III

58) **Statement-I :** GM crops has more tolerance to abiotic stress.
Statement-II : All strains of *Bacillus thuringiensis* produces protein that kill certain insects.

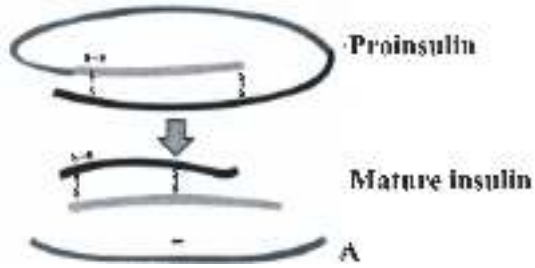
(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

59) Given figure represents the maturation of pro-insulin into insulin. Identify the product A.



(1) Polypeptide chain A

(2) Polypeptide chain B

(3) Polypeptide chain C

(4) Polypeptide chain D

60) Baculoviruses used as biological control agents belong to :

(1) Genus *Glomus*

(2) Genus *Nucleopolyhedro* virus

(3) Genus *Escherichia*

(4) Genus *Bacillus*

61) Consider the following statements (a – d)
(a) Roquefort cheese are ripened by growing a specific bacteria on them
(b) Different varieties of cheese are known by their characteristic texture, flavour and taste
(c) Statins act by competitively inhibiting the enzyme responsible for synthesis of cholesterol
(d) Methanogens are commonly found in aerobic sludge during sewage treatment
Select the option which includes incorrect statements only

(1) a, b and c only

(2) only d

(3) a and d only

(4) b and c only

62) Match column-I with column-II and select the correct option using the codes given below :

Column-I		**Column-II**	
(a)	Dragonflies	(i)	Attack insects and Arthropods
(b)	Baculovirus	(ii)	Biogas production
(c)	*Glomus*	(iii)	Get rid of mosquitos
(d)	Methanogens	(iv)	Absorb phosphorus from soil

(1) a - (iii), b - (i), c - (iv), d - (ii)

(2) a - (ii), b - (i), c - (iii), d - (iv)

(3) a - (iii), b - (iv), c - (ii), d - (i)

(4) a - (i), b - (iii), c - (iv), d - (ii)

63) Which of the following bacteria was associated with discovery of penicillin ?

(1) *Propionibacterium*
(2) *Trichoderma*
(3) *Streptococcus*
(4) *Staphylococcus*

64) Biogas consists of

(1) Carbon monooxide, methane and hydrogen
(2) Carbon dioxide, methane and hydrogen
(3) Carbon monooxide, ethane and hydrogen
(4) Carbon dioxide, ethane and hydrogen

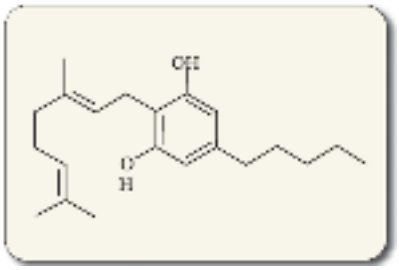
65) Saliva in mouth is :

(1) Physical barrier
(2) Physiological barrier
(3) Cellular barrier
(4) Cytokine barrier

66) Which of the following are ionising radiation that act as carcinogen?

(1) x-rays
(2) UV rays
(3) gamma rays
(4) both 1 and 3

67) Identify structure



(1) Opioids
(2) Cannabinoid
(3) Morphine
(4) Heroine

68) Chemical name of smack is :-

(1) Amphetamine
(2) Diacetyl morphine
(3) Triacetyl morphine
(4) Diacetyl alcohol

69) Which disease is caused by rhinoviruses ?

(1) AIDS
(2) Typhoid
(3) Pneumonia
(4) Common cold

70) Which stage of Plasmodium is introduced into the human body through the bite of an infected mosquito ?

(1) Sporozoites
(2) Merozoites
(3) Trophozoites
(4) Gametocytes

71) **Assertion :** Transplantation of tissue from one body part into another body part of an individual is called autograft.
Reason : Heterograft is the most successful type of tissue grafting.

(1) Both assertion and reason are true and the reason is not a correct explanation of the assertion.
(2) Assertion is true but the reason is false
(3) Assertion is false but the reason is true
(4) Both assertion and reason are true and the reason is a correct explanation of the assertion

72) Select correct option about malaria.

		Human	**Mosquito**
(1)	Infected stage of plasmodium	Gametocytes	Sponrozoites
(2)	Types of reproduction	Asexual	Asexual
(3)	Infected stage of plasmodium	Sporozoites	Gametocytes
(4)	Types of reproduction	Sexual	Asexual

(1) 1
(2) 2
(3) 3
(4) 4

73) Match the column-

	Column-I		**Column-II**
(A)	Ionisation radiation	(I)	UV Rays
(B)	Non ionizing radiation	(II)	x-rays & γ-rays
(C)	Viral oncogene	(III)	C-onc
(D)	Cellular oncogene	(IV)	Oncogenic virus

(1) A–I, B–II, C–III, D–IV

(2) A–IV, B–III, C–II, D–I

(3) A–II, B–I, C–IV, D–III

(4) A–II, B–I, C–III, D–IV

74) Read the given statements carefully and select the correct option.
Statement-I :- We still do not understand how higher vertebrates have the ability to differentiate pathogens from self-cells.
Statement-II :- In auto immune disease, the body attack self-cells.

(1) Statement I and statement II both are incorrect

(2) Statement I is incorrect but statement II is correct

(3) Statement I and statement II both are correct

(4) Statement I is correct and statement II is incorrect

75) Read the given statements carefully and select correct option.
Statement-I :- Immunotherapy is a treatment procedure that involves suppression or augmentation of immune response to achieve therapeutic effects.
Statement-II :- Immunosuppression is required in case of organ transplantation.

(1) Statement II is correct but Statement I is incorrect

(2) Both statement I and Statement II are correct

(3) Both Statement I and Statement II are incorrect

(4) Statement I is correct but Statement II is incorrect

76) Which are of the following is not related to withdrawal syndrome ?

(1) Observed, when regular dose of drugs/alcohol is discontinued.

(2) Symptoms may be relieved when use is resumed again.

(3) Severe and life saving

(4) Patient may need medical supervision

77) Following statements are given below. How many statements are correct ?
(a) When pathogen enters first time inside the body. body immune system rapidly recognise the pathogen and responds quickly to it.
(b) Active immunity is slow and takes time to given its its full effective response.
(c) The foetus receives some antibodies from their mother, through the placenta during pregnancy, these is example of Active immunity.
(d) Secondary immune response is based on memory of immune system.

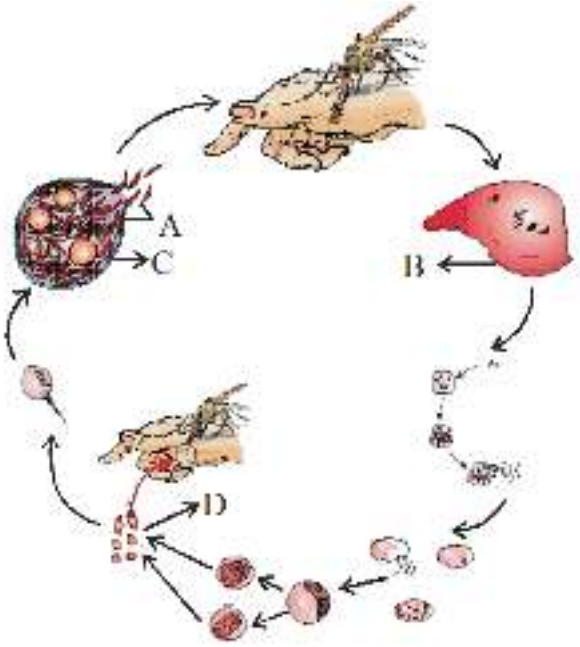
(1) Two

(2) Three

(3) Four

(4) One

78) Identify the likely A, B, C, D in the following life cycle.



	A	B	C	D
(1)	Gametocyte	Salivary gland	Liver	Sporozoites
(2)	Sporozoites	Salivary gland	Liver	Gametocyte
(3)	Sporozoites	Liver	Salivary gland	Gametocyte
(4)	Sporozoites	Liver	Gametocyte	Salivary gland

(1) 1

(2) 2

(3) 3

(4) 4

79) Which of the following common causes, motivates youngsters towards drug & alcohol use?
(A) Curiosity
(B) Need for adventure
(C) Experimentation
(D) Adolescence

(1) (A), (B), (D)

(2) (B), (C), (D)

(3) (A), (B), (C)

(4) (A), (C), (D)

80) **Assertion :** Adaptive rediation leads to the evolution of different species from a common ancestor in a particular gregraphical area.
Reason : different species evolve simultaneously in different habitats due to the same selection pressure.

(1) Both assertion and reason are true but reason is not the correct explanation of assertion.

(2) Both assertion and reason are true and reason is the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

81) Mark the incorrect statement

(1) Two million years ago, australopithecines probably lived in eastern Africa.

(2) Homo habilis was with a brain size of 650-800 cc

(3) Homo erectus had a large brain around 1400 cc

(4) The agriculture come around 10000 years back and human settlements started.

82) **Statement-I :-** Ontogeny is the study of embryological development of individual organisms.
Statement-II :- Recapitulation theory explained by earnst haeckel.

(1) Only Statement-I is correct.

(2) Statement-I and II both are incorrect.

(3) Statement-I and II both are correct.

(4) Only Statement-II is correct.

83) Match the column -

	Column-I		**Column-II**
(i)	*Ramapithecus*	(A)	650-800 cc Brain capacities
(ii)	*Homo habilis*	(B)	ate meat
(iii)	*Homo erectus*	(C)	More man like

Which of the following is correctly match?

(1) (i)-B, (ii)-A, (iii)-C

(2) (i)-C, (ii)-A, (iii)-B

(3) (i)-C, (ii)-B, (iii)-A

(4) (i)-B, (ii)-C, (iii)-A

84) **Assertion:** Louis pasteur by carful experimentation demonstrated that life comes only from pre-existing life. He showed that in pre-sterilized swan neck flasks, life did not come from killed yeast.
Reason: First form of life comes from pre-existing living organic molecules during chemical evolution.

(1) Assertion and Reason both are correct and R is correct explanation of A

(2) A and R are true but R is not correct explanation of A

(3) A is true but R if false

(4) A is false and R is true

85) Which of the following statements are correct about cancer?
A. Cancer cells can spread from the primary tumour to distant organs through blood or lymph, a process called metastasis, which is responsible for the malignant nature of cancers.
B. Ionising radiations such as X-rays and γ-rays can cause permanent damage to DNA increasing

mutation rates and thereby raising the risk of cancer development.
C. All cancers are caused exclusively by viral infections; environmental factors do not contribute.
D. DNA damage, increases the risk of tumor formation.

(1) A, B, and D only
(2) A and C only
(3) B, C, and D only
(4) A. B, C, and D

86) Which of the following statement is incorrect ?

(1) Stellar distances are measured in light years.
(2) There was no atmosphere on early earth.
(3) Life appeared 0.5 billion years after the formation of earth.
(4) The first non-cellular forms of life could have originated 2 billion years back

87) In a natural selection two peaks are formed. This is an example of which type of natural selection ?

(1) Stabilizing
(2) Directional
(3) Disruptive
(4) Progressive

88) Which can be used as industrial pollution indicators?

(1) Moths
(2) Lichens
(3) Rabbit
(4) Rat

89) __A__ support for evolution was proposed by Ernst Heckel.
Fill in the blanks with correct option :

(1) Embryological
(2) Palaeontological
(3) Biochemical
(4) Biogeographical

90) Gene therapy performed for correction of disorders :-

(1) Is always a temporary cure
(2) Could be permanent if gene is introduced in the early embryonic stages
(3) Has not been tried for humans
(4) Involves removal of functional genes & introduction of defective genes

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	4	4	3	1	1	2	4	1	2	4	4	3	1	2	1	2	2	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	4	4	3	1	1	4	3	2	3	2	4	4	2	4	1	1	4	2	2
Q.	41	42	43	44	45															
A.	3	1	2	3	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	1	2	2	3	1	4	1	2	2	2	2	4	1	3	2	1	4	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	1	3	3	1	3	2	2	3	4	4	1	2	4	4	1	1	1	4
Q.	86	87	88	89	90															
A.	3	2	1	4	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	2	4	4	4	1	2	1	3	3	3	2	2	4	4	2	1	4	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	2	3	2	4	3	1	4	3	1	4	1	2	3	1	4	3	2	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	4	3	2	1	4	1	3	3	4	1	4	3	2	3	2	3	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	4	2	2	4	2	2	4	1	2	3	3	3	2	3	1	3	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	2	3	1	4	3	2	1	2										

SOLUTIONS

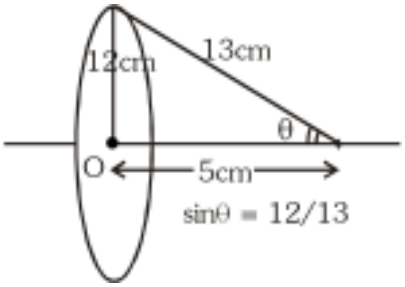
PHYSICS

1) $B = \frac{\mu_0 i}{2\pi r} = \frac{2 \times 10^{-7} \times 35}{0.20}$

$= 3.5 \times 10^{-5}$ T

2) $B = B^{0} \sin^3\theta = 0.5 \times 10^{-4} \left(\frac{12}{13}\right)^3$

$= 3.9 \times 10^{-5}$ T



3) $B_{net} = \frac{\mu_0 i}{2r} - \frac{\mu_0 i}{2(2r)} + \frac{\mu_0 i}{2(2^2 r)} \ldots\ldots\infty$

$B_{net} = \frac{\mu_0 i}{2r}\left[1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \ldots\ldots\ldots\infty\right]$

The term in the bracket in an infinite geometric progression with first term a = 1 and common ratio $r' = \frac{-1}{2}$

Sum of infinite GP $S_\infty = \frac{a}{1 - r'}$

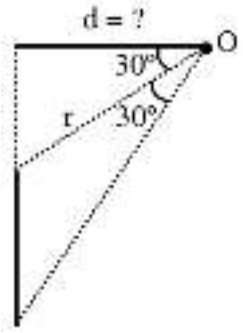
$S_\infty = \frac{1}{1 + \frac{1}{2}} = \frac{2}{3}$

$B_{net} = \frac{\mu_0 i}{2r} \times \frac{2}{3}$

$B_{net} = \frac{\mu_0 i}{3r}$

4) $d = r\cos 30° \Rightarrow d = \frac{\sqrt{3}r}{2}$

$\theta_1 = 60°$; $\theta_2 = -30°$



$$B_0 = \frac{\mu_0 I}{4\pi d}\left[\sin 60^\circ - \sin 30^\circ\right]$$
$$= \frac{\mu_0 I}{4\pi \frac{\sqrt{3}r}{2}}\left[\frac{\sqrt{3}-1}{2}\right]$$
$$B_0 = \frac{\mu_0 I}{2\sqrt{3}\pi r}\left[\frac{\sqrt{3}-1}{2}\right]$$

5) A magnetic field exerts a force perpendicular to the velocity of a charged particle, altering its direction (momentum) but not its speed (kinetic energy).

6) $B = \mu_0 ni$

$$= 4\pi \times 10^{-7} \times \frac{5 \times 400}{80 \times 10^{-2}} \times 8$$
$$= 8\pi \times 10^{-3}\ T$$

7) Mean radius,

$$r_m = \frac{0.25 + 0.26}{2} = 25.5 \times 10^{-2} m$$
$$n = \frac{N}{2\pi r_m} = \frac{3500}{2\pi \times 25.5 \times 10^{-2}}$$

Magnetic field of a toroid

$B = \mu_0 nI$

$$= 4\pi \times 10^{-7} \times \frac{3500}{2\pi \times 25.5 \times 10^{-2}} \times 11$$
$$= 3 \times 10^{-2} T$$

8) By using $i = \frac{e}{R} = -\frac{1}{R}\frac{d\phi}{dt}$

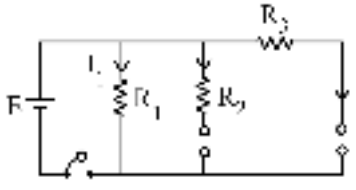
$$i = -\frac{1}{10}\frac{d}{dt}\left(6t^2 - 5t + 1\right) = -\frac{1}{10}(12t - 5)$$
$$i = -\frac{1}{10}(12 \times 0.25 - 5) = 0.2A$$

9) $F_M = \frac{B^2 l^2 v}{R} = \frac{(0.15)^2 \times (0.5)^2 \times 2}{3}$

$$= 3.75 \times 10^{-3} N$$

10)

At t = 0, inductor behave as an open circuit



$I_1 = \frac{E}{R_1}$, $I_2 = I_3 = 0$

11) e.m.f. = $(\vec{v} \times \vec{B}) \cdot \vec{\ell}$ and high potential side will be in the direction of $\vec{v} \times \vec{B}$

12)

Potential difference across the two ends :-

$e = \left(\vec{v} \times \vec{B}\right) . \vec{\ell}$

when any two vector are parallel then induced e.m.f will be zero

13) This is the case of Resonant condition

V = iR

100 = i × 50

i = 2A

14) $V = \sqrt{V_R^2 + V_C^2}$

$10 = \sqrt{V_R^2 + 8^2}$

$\Rightarrow V_R = 6V$

$\tan\phi = \frac{V_C}{V_R} = \frac{8}{6} = \frac{4}{3}$

15) Displacement current = conduction current

$\frac{dq}{dt} = \frac{d}{dt}[q^0 \cos 2\pi\upsilon t] = - q^0 2\pi\upsilon \sin 2\pi\upsilon t$

16) Given,

$\vec{B} = 3 \times 10^{-8} \sin (1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{j} T$

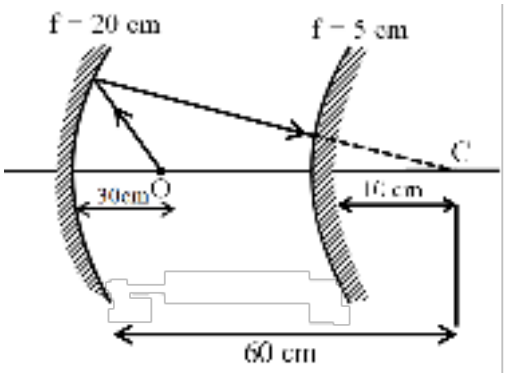
Here $B_0 = 3 \times 10^{-8}$ T;

Also, $\frac{E_0}{B_0} = c \Rightarrow E^0 = B^0 \times c$

$E_0 = 3 \times 10^{-8} \times 3 \times 10^8 = 9$ NC^{-1}

Hence, $\vec{E} = 9 \sin (1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{k}$ Vm^{-1}

17)



x : 60 – 10 = 50 cm

18) d = 30°
A = 60° 30° = i + e – 60°
i + e = 90° let i = r = 45° ⇒ d_{min} = 30°

$$\mu = \frac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin A/2} = \frac{\sin 45^0}{\sin 30^0}$$

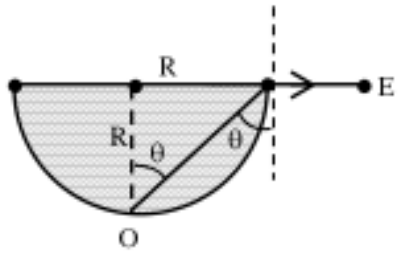
$$= \frac{1/\sqrt{2}}{1/2} = \sqrt{2}$$

By increasing μ, δ_{min} increases hence, deviation cannot be 30°. Hence (2)

19) $$\frac{dv}{du} = -\left(\frac{f}{u-f}\right)^2 \quad dv = -b\left(\frac{f}{u-f}\right)^2$$

20) Refractive index has no relation with mass density because both have different meaning. Hence reason is incorrect.
So (A) is correct but (R) is not correct.

21)



$$\sin c = \frac{1}{\mu}$$

for μ → least,
c → maximum
θ = c = 45°

$$\mu = \frac{1}{\sin C}$$

$$= \frac{1}{\sin 45^\circ}$$

$$\mu = \sqrt{2}$$

22) $$\frac{1}{f_1} = (1.75-1)\left(-\frac{1}{30}\right)$$ ⇒ f_1 = –40 cm

$$\frac{1}{f_2} = (1.75-1)\left(\frac{1}{30}\right) \Rightarrow f_2 = 40\text{cm}$$

Image from L_1 will be virtual and on the left of L_1 at focal length 40 cm. So the object for L_2 will be 80 cm from L_2 which is 2f. Final image is formed at 80 cm from L_2 on the right.
So x = 120

23) $$\frac{\Delta}{\lambda} = \frac{\phi}{2\pi}$$

$$\frac{\lambda}{6\lambda} = \frac{\phi}{2\pi} \Rightarrow \phi = \frac{\pi}{3}$$

$I = I^0 \cos^2\frac{\phi}{2}$

$I = I^0\cos^2\frac{\pi}{6}$

$\frac{I}{I_0} = \frac{3}{4}$

24) For brightness, path difference = $n\lambda = 2\lambda$
So second is bright.

25)

When unpolarised light is incident on a medium of refractive index 1.73 at Brewster's angle then reflected light is completely polarised.
Using Brewster's Law : $\mu = \tan(i_p)$
$\Rightarrow 1.73 = \tan(i_p) \Rightarrow i_p = 60°$
By Law of reflection, $i_p = r = 60°$

26)

$\sin 30° = \frac{2\lambda}{a}$

$\Rightarrow \lambda = \frac{24 \times 10^{-5}}{4} = 6 \times 10^{-5}\text{cm}$

or $\lambda = 6000$ Å

27) **Solution :**
$E = \Delta m.C^2$

$\Rightarrow E = \frac{0.3}{1000} \times \left(3 \times 10^8\right)^2 = 2.7 \times 10^{13}\,\text{J}$

$= \frac{2.7 \times 10^{13}}{3.6 \times 10^6} = 7.5 \times 10^6\,\text{kWh}$

28) As the collision is inelastic, it means a part of K.E. is transformed into some other form due to collision. In this case, the KE of incident electron can be absorbed by H atom and it can absorb only 10.2 eV out of 11.2 eV, so that it can reach to 1st excited state and electron leaves with remaining energy i.e. 1.0 eV.

29)

Radius (r) $\propto \frac{n^2}{Z}$, velocity (v) $\propto \frac{Z}{n}$
(P) Magnetic dipole moment (μ)
$\mu = IA$ [I → Current and A → Area of the orbit]
$I = \frac{e}{T} = \frac{eV}{2\pi r} \propto \frac{V}{r} \propto \frac{Z/n}{n^2/Z} \propto \frac{Z^2}{n^3}$

Area (A) = $\pi r^2 \propto \left(\frac{n^2}{Z}\right)^2 \propto \frac{n^4}{Z^2}$

$\mu \propto \left(\frac{Z^2}{n^3}\right)\left(\frac{n^4}{Z^2}\right) \propto n^1 Z^0$

(P) → 1, 4

(Q) Magnetic field at the centre (B)

The magnetic field at the centre of a circular current loop is $B = \frac{\mu_o I}{2r}$

$$B \propto \frac{1}{r} \propto \frac{Z^2/n^3}{n^2/Z} \propto \frac{Z^3}{n^5}$$

(Q) → 3

(R) Angular velocity

$\omega = \frac{v}{r}$

$\omega \propto \frac{v}{r} \propto \frac{Z/n}{n^2/Z} \propto \frac{Z^2}{n^3}$

(R) → 2, 5

(S) Potential energy in a Bohr atom-

$U = -\frac{kZe^2}{r}$

$U \propto \frac{Z}{r} \propto \frac{Z}{n^2/Z} \propto \frac{Z^2}{n^2}$

(S) → 2, 6

30) Energy of photon

$= \frac{hc}{\lambda} = hcR\left(\frac{1}{2^2} - \frac{1}{5^2}\right) = \frac{21hcR}{100}$

Momentum of photon = $\frac{E}{c} = \frac{21hR}{100}$

=Momentum of atom

Velocity of atom = $\frac{21hR}{100m}$ (where m = mass of atom)

31) De Broglie wavelength of a particle, $\lambda = \frac{h}{p}$

Where p is the momentum of the particle and h is the planck's constant.

Kinetic energy of the particle, $K = \frac{p^2}{2m}$

Where m is the mass of the particle

$\therefore \quad \lambda = \frac{h}{\sqrt{2mK}}$

Since mass of the particle remains constant

$\lambda \propto \frac{1}{\sqrt{K}}$

$\frac{\lambda'}{\lambda} = \sqrt{\frac{K}{K'}} = \sqrt{\frac{1eV}{1 \times 10^6 eV}} = \frac{1}{10^3}$

$\lambda' = \frac{\lambda}{10^3} = \frac{2000}{10^3}\text{Å} = 2\text{Å}.$

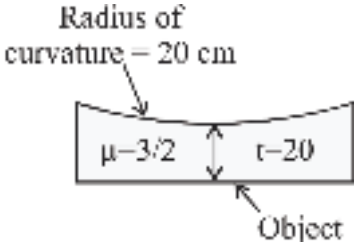
32) Due to motion of majority carriers from high concentration to low concentration, diffusion current is established whose direction is from P to N region.

33) $Y = \overline{\overline{A+B}}.B$
$= (A + B) . B = A . B + B . B$
$= A . B + B = B (A + 1) = B$

34) In zener diode covalent bond in depletion layer break due to electric field so doping high used and width of deplection layer is thin.

35) $Q = \frac{\omega}{\Delta\omega} = \frac{\omega L}{R} \Rightarrow \Delta\omega = \frac{R}{L} = \frac{80}{8} = 10$ Hz
$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 50 \times 10^{-6}}} = 50$
$\omega_L = \omega - \frac{\Delta\omega}{2} = 50 - 5 = 45$ rad/s
$\omega_H = \omega + \frac{\Delta\omega}{2} = 50 + 5 = 55$ rad/s

36) Radius of curvature = 20 cm



Considering refraction at the curved surface,
$u = -20$; $\mu_2 = 1$
$\mu_1 = 3/2$; $R = +20$
Applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$
$\frac{1}{v} - \frac{3/2}{-20} = \frac{1-3/2}{20} = v = -10$
i.e. 10 cm below the curved surface or 10 cm above the actual position of flower.

37)

$\phi_{Ph} = \phi_0 + KE_{max}$
$\Rightarrow \phi_{Ph} = h\upsilon_0 + KE_{max}$
$\Rightarrow 8$ eV $= 10.56 \times 10^{-19}$ Joule $+ KE_{max}$
$\Rightarrow 8$ eV $= \frac{10.56 \times 10^{-19}}{1.6 \times 10^{-19}}$ eV $+ KE_{max}$
$\Rightarrow 8$ eV $= 6.6$ eV $+ KE_{max}$

$\Rightarrow KE_{max} = 1.4$ eV

38)

The energy of the incident photons is given by $E = \frac{1240}{\lambda} eV$.

For metal p: $E_p = \frac{1240}{560} \approx 2.21$ eV.

Stopping potential for p: $eV_{0p} = 2.21 - 2.0 = 0.21$ eV $\Rightarrow V_{0p} = 0.21$ V.

For metal $E_q = \frac{1240}{450} \approx 2.75$ eV.

Stopping potential for q: $eV_{0q} = 2.75 - 2.5 = 0.25$ eV $\Rightarrow V_{0q} = 0.25$ V.

For metal r: $E_r = \frac{1240}{350} \approx 3.54$ eV.

Stopping potential for r: $eV_{0r} = 3.54 - 3.0 = 0.54$ eV $\Rightarrow V_{0r} = 0.54$ V.

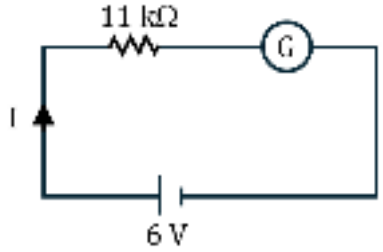
The magnitude of stopping potentials follows the order: $|V_{0r}| > |V_{0q}| > |V_{0p}|$.

On the voltage axis the intercept for r will be furthest to the left.

Since all beams have equal intensities the saturation current for all then be the same.

Graph 4 correctly shows r having the highest stopping potential and all curves reaching the same saturation current.

39)

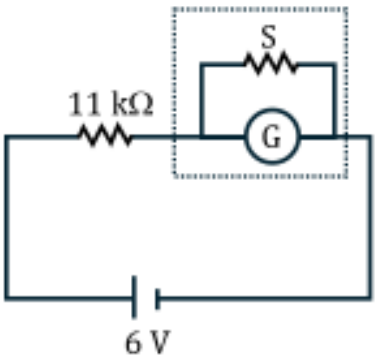


$I = 9 \times 60\ \mu A = 540 \times 10^{-6}$ A

$$\frac{6}{11000 + G} = 540 \times 10^{-6}$$

$$G = \frac{1000}{9}$$

If shunt is applied



From the standard relation from half deflection method.

$$G = \frac{RS}{R - S} \Rightarrow \frac{1000}{9} = \frac{11000\,S}{11000 - S}$$

$S = 110\ \Omega$

40)

$i + e = \delta + A$
$A = 45°$

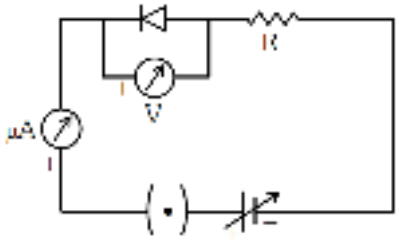
41)

Real depth = (23.25 – 5.25) cm = 18.00 cm
Apparent depth
= (23.25 – 11.25) cm = 12.00 cm

$$RI = \frac{\text{Real depth}}{\text{App. depth}} = \frac{18.00}{12.00} = 1.5$$

42)

Why circuit (1) implements this correctly -
The diode is reverse-based its cathode is corrected to the positive terminal of the variable source, anode to the negative.
The series resistor R and the µA -meter is placed in series with the diode so every electron passing through the diode also passes through the meter.
The voltmeter is strapped directly across the diode's across the diodes terminals so it reads only V_D not the drop across R.
By gradually increasing the reverse bias V, one can record pairs (V_D, I) and plot the reverse-bias V-I curve.



Final answer: Option (1)

43)

$I = I_Z + I_L$

$$I = \frac{9-6}{100} = 30\,\text{mA}$$

$$I_L = \frac{6\text{V}}{1\,\text{k}\Omega} = 6\,\text{mA}$$

$I_Z = I - I_L = 30 - 6 = 24$ mA

44)

(a) Diode is undirectional
(b) When capacitor is fully charged, current through it be zero
(c) LED emits light
(d) Carbon resistor having several rings

45) $M = \frac{-f_0}{f_e}$, $L = |f^0| - |f_e|$

$50 = \frac{-100}{f_e}$

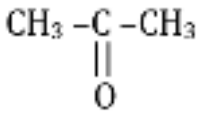
$f_e = -2$ cm

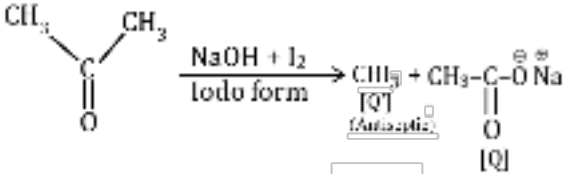
L = 100 – 2 = 98 cm

CHEMISTRY

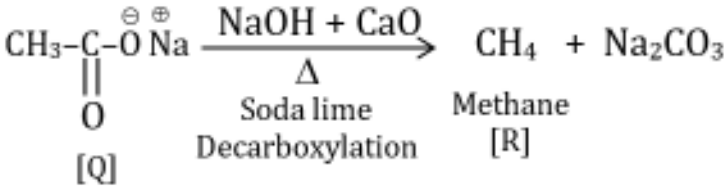
46)

Solution:

First member of ketone family is  : propanone



Q' is antiseptic



 Here formation of 'R' involves formation of carbanion intermediate.

Final Answer: option (4)

47)

Reactivity towards $SN^2 \propto \frac{1}{\text{sterichindrance}}$

48) **Solution:**

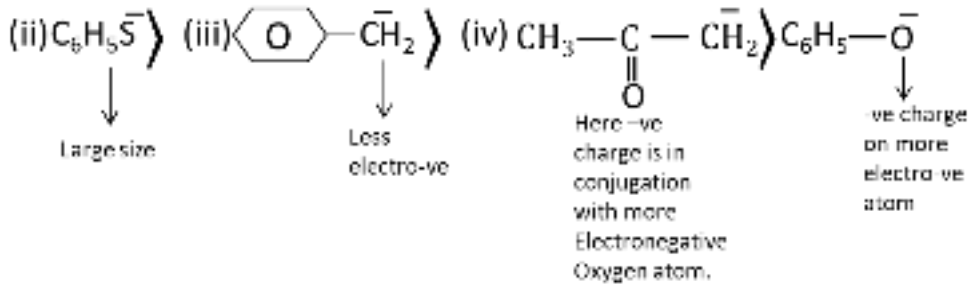
•One which donates its electrons readily are said to be good nucleophiles and has more Nucleophilicity.

•If lone pair is in conjugation then tendency to donate electron decreases and Nucleophilicity also decreases.

•Nucleophilicity a size [along the group]

•Nucleophilicity $\propto \frac{1}{\text{Electronegativity}}$ [along the period]

∴ Order of Nucleophilicity is :



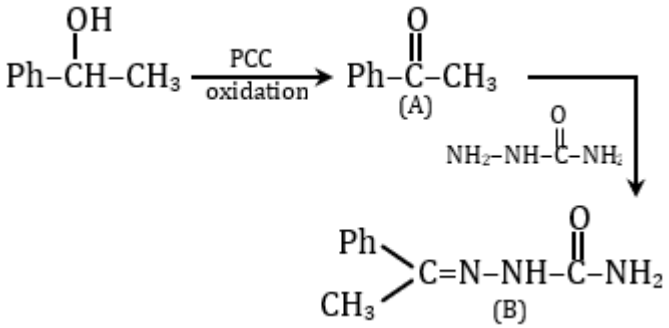
Final answer: Option (1)

49)

All are reduction Reaction and products are correctly match according to Reactants and Reagents.

50)

Explain the Question :- Find out the final product 'B' in the given reaction.
Concept :-This question is based on oxidation, NAR (addition of semicarboazide.)
Solution :-



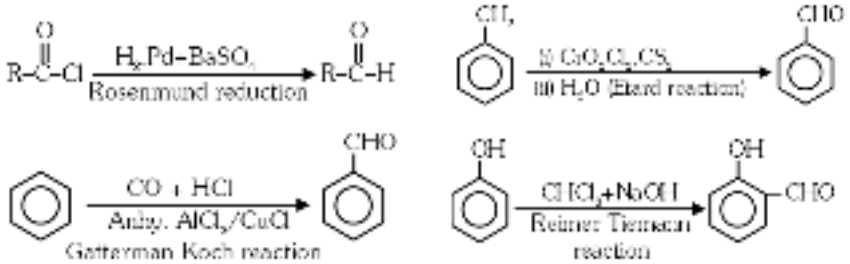
Final answer :- (2)

51)

Reaction belongs to ammonia derivatives.

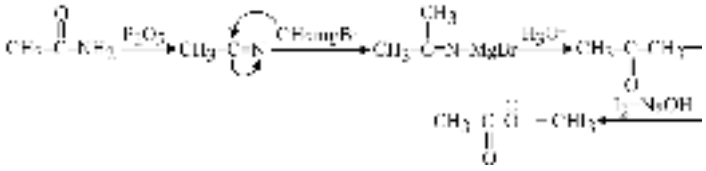
52)

Solution :



Correct matching: A-IV, B-III, C-I, D-II

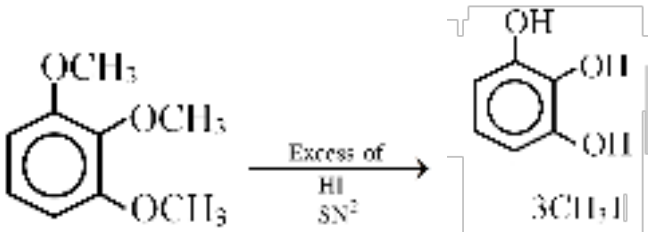
Final answer : 1

53) 

54) **Solution:**

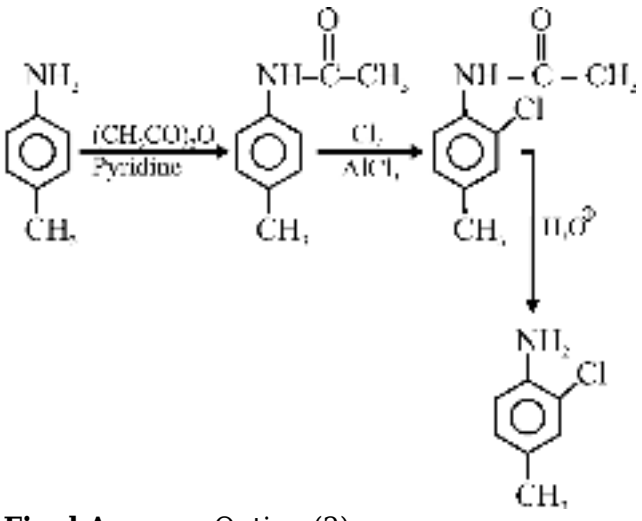
Reaction (Column-I)	Reagent / Condition	Key Reaction	Product Type (Column-II)
A. CH_3COOH	P_2O_5 , Δ	Dehydration of acid	Anhydride (Q)
B. Salicylic acid	$(CH_3CO)_2O/H^+$	Acetylation	Analgesic - Aspirin (P)
C. $CH_3COOC_2H_5$	DIBAL-H	Partial reduction of ester	Aldehyde (S)
D. Ketone	NH_2NH_2/KOH/Ethylene	Wolff-Kishner reduction	Alkane (R)

Final Answer: The correct option is (1).

55) 

56)

Solution:



Final Answer: Option (2)

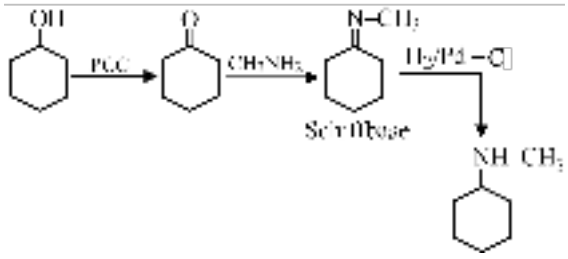
57)

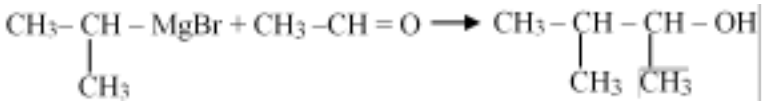
Products are according to nature of reactants $NaBH_4$, H_2/pt

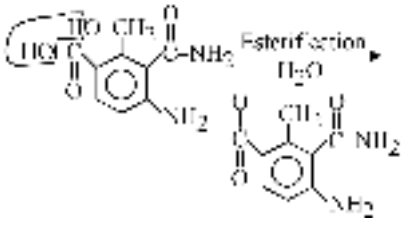
58)

Both are correct

59) **Solution :**
Statement 1 is correct : Nucleotides are joined by phosphodiester bonds between the 5' and 3' carbons of the sugar.
Statement 2 is correct : DNA strands are complementary due to specific hydrogen bonding between base pairs.
Statement 3 is correct : DNA replicates during cell division, ensuring identical genetic material is passed to daughter cells.
Final answer : 4

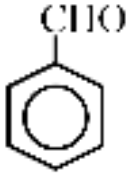
60) 

61) 

62) 

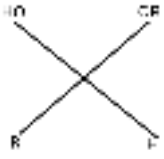
63)

H–CHO gives only canizaro



do not involve in haloform test

64)

⇒ An any with a hemiacetal  are is equilibrium with a ring opened form containing an aldehyde and will react as a reducing agent towards certain (oxidizing) metal salts such as Cu^{2+} (Benedict's test, Fehling's solution) and Ag^{+} (tollen's test) are there fore called as reducing sugar.
⇒ Since all sugar are with hemiacetal group, hence all are reducing sugars.

Final Answer : Option: (4)

65)

A) Question Explanation: The question asks to correctly match methods/tests with their outcomes or purification techniques.
B) Concept Used: Chromatography, Lassaigne's test for N and S, Steam distillation, Vacuum distillation
C) Answer Explanation: Main Rule: Match each method with its characteristic result.

Column-I Item	Column-II Match	Reason
(A) Best and latest technique	(i) Chromatography	Modern separation method
(B) Lassaigne's test for N	(iv) Prussian blue coloured complex	NaCN + Fe → Prussian blue
(C) Lassaigne's test for S	(iii) Violet coloured complex	Na_2S + Nitroprusside → Violet
(D) Aniline purified from water	(v) Steam distillation	Aniline is steam volatile
(E) Glycerol purified from spent lye	(ii) Vacuum distillation	Glycerol decomposes at bp

The Correct matching order is (A)-(i), (B)-(iv), (C)-(iii), (D)-(v), (E)-(ii). This matches Option 3.D)
Final Answer: Option (3)

66) **Explain the Question :** We need to find the nitrogen percentage in a 0.75 g soil sample. In the Kjeldahl method, the nitrogen is converted on NH_3, which is then neutralized by 10 ml of 1M H_2SO_4.
Concept : This question is based on the formula for the percentage of nitrogen is

$$\%N = \frac{1.4 \times N \times V}{w}$$

N (Normality of H_2SO_4) : Since H_2SO_4 is dibasic, N = M × 2 = 1 × 2 = 2N.
V (Volume) : 10 ml
w (Weight of sample) : 0.75g
Solution :
Plug the values into the formula :

$$\%N = \frac{1.4 \times 2 \times 10}{0.75}$$

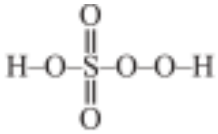
$$\%N = \frac{28}{0.75}$$

%N = 37.33
Final Answer: (1)

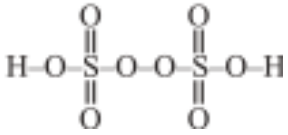
67)

It's an example of capacity Reaction by which we form azodyes

H_2SO_5 (Caro's acid)



$H_2S_2O_8$ (Marshall's acid)



68)

69) $NaOH + Zn \rightarrow NaZnO_2 + H_2$

70)

Explaining the Question :-

Question is asking about group reagent of basic radicals in group 4

Concept:-

group reagent of basic radicals

Solution :-

$NH_4OH + H_2S$ gas is used as group reagent of basic radicals
Final Answer :- 3

71)

Borax bead test is given by coloured ions of d-block elements ie, Cu^{2+}

72) **Solution :-** The ion that gives a **red precipitate** on addition of **dimethylglyoxime (DMG)** is Ni^{2+} (nickel(II)).
Final Answer :- 3

73) **Explaining the Question :-**
Question is asking about the compound that do not give chromyl chloride test.
Concept:-
chromyl chloride test
Solution :-
$HgCl_2$ do not have chromyl chloride test because $HgCl_2$ is made up of a Hg^{2+} ion (a dimer of mercury(I) ions), where two Hg^{+} ions are bonded together via a chloride ion.
The Hg^{+2} ion has a +2 oxidation state, not +2 or +3 like chromium(III) in the cromyl chloride test.
$HgCl_2$ does not contain the necessary Cr^{3+} ions required to form cromyl chloride (CrO_2Cl_2)
Final Answer :- 2

74) Blue lake test is given by Al^{+3}. It is based on adsorption principle where blue litmus solution is absorbed by a electrons aluminium hydroxide precipitate, creating a coloured "lake" that floats on the solution.

Hence,

Option (2) is correct : Al^{3+}

75) $\underset{(A)}{CaCl_2} \xrightarrow{K_2Cr_2O_7/H^+} \underset{(B)}{CrO_2Cl_2} \xrightarrow{NaOH} \underset{(C)}{Na_2CrO_4} \xrightarrow{(CH_3COO)_2Pb} \underset{(D)}{PbCrO_4}$

76) $NO_2^- + \text{dil. } H_2SO_4 \rightarrow NO_2 \uparrow$

Brown

$Br^- + \text{Conc. } H_2SO_4 \rightarrow Br_2 \uparrow + SO_2 \uparrow$

Brown coloured gas

77) Fe^{+2} + $[Fe(CN)_6]^{-4}$ → White ppt.
Fe^{+3} + $[Fe(CN)_6]^{-4}$ → Prussian blue ppt.
Zn^{+2} + $[Fe(CN)_6]^{-4}$ → Bluish white ppt.
Ag^{+2} + $[Fe(CN)_6]$ → White ppt.
Cu^{+2} + $[Fe(CN)_6]$ → Chocolate brown ppt.
Ca^{+2} + $K_4[Fe(CN)_6]$ → White ppt.

78) "MnO_2" has less oxidising action F_2. Hence F^- can't be oxidised by MnO_2.

79) Concept

80)

Explanation – The question asks you to identify the chemical reaction where the central atom's hybridization state remains unchanged from reactants to products.

Concept (Hybridization & Hydrolysis reaction)

PCl_5 (sp^3d) + $5H_2O$ → H_3PO_4 (sp^3) + 5HCl (Hybridization change)

SF_4 (sp^3d) + $4H_2O$ → H_2SO_3 (sp^3) + 4HF (Hybridization change)

XeF_6 (sp^3d^3) + $6H_2O$ → XeO_3 (sp^3) + 6HF (Hybridization change)

$SiCl_4$ (sp^3) + $4H_2O$ → $Si(OH)_4$ (sp^3) + 4HCl (No hybridization change)

Final Answer – Option(4)

81) Weak metallic bond of Mn

82) Osmium show +8 O.S. Manganese show max +7 oxidation state.

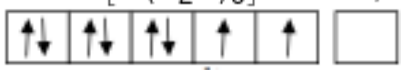
83) Alloy formation → Similar radii

84)

$Yb^{+2} = [Xe]4f^{14}$
$Lu^{+3} = [Xe]4f^{14}$

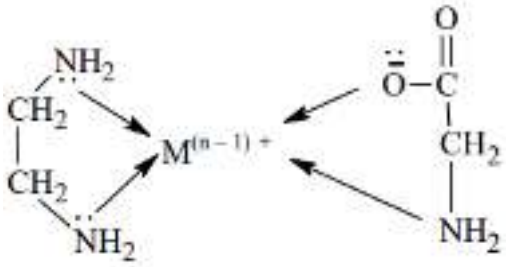
85) '3d' subshell does not contain unpaired electron.

86) (1) The magnitude of splitting energy increases down the group. Thus
$Pt^{2+} - 5d^8, Pd^{2+} - 4d^8$ and $Ni^{2+} - 3d^8$.
(2) NH_3 is a borderline ligand which forces pairing of electrons in Co^{3+}, whereas H_2O is a weak-field ligand and it can not force the pairing in Ni^{2+}.
Ni^{2+} in $[Ni(H_2O)_6]^{2+} - 3d^8, 4s^0$



So, there is no unpaired electron in $[Zn(NH_3)_4]^{+2}$ whereas there are unpaired electron in $[Ni(H_2O)_6]^{2+}$. Thus $[Zn(NH_3)_4]^{+2}$ is colourless and $[Ni(H_2O)_6]^{2+}$ is coloured due to d–d transition.

(3)



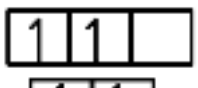
As en is symmetrical ligand thus there is no geometrical isomerism.
(4) In $K_3[Fe(CN)_6]$, the oxidation state of Fe is +3.
$[Fe(CN)_6]^{3-} - 3d^5, 4s^0$

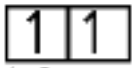


Number of unpaired electron = 1
$\therefore \mu = \sqrt{n(n+2)}$ B.M.
$= \sqrt{1(1+2)}$ B.M. $= \sqrt{3}$ B.M.

87) SFL present , Soil forms I.O.C (Mn^{+3} complex) remaining W.F.L.

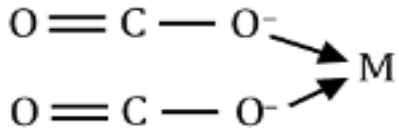
88) CFSE = Value for $3d^4$ (Tetrahedral complex)





$\left(-\frac{3}{5} \times 2 + \frac{2}{5} \times 2\right) \Delta t \Rightarrow -\frac{2}{5} \Delta_t$

89) **Question is Asking About:**
In this question we have to find out bidentate ligand among given.
Concept:
When a ligand has two groups or atom that are capable of bonding to the central metal atom. it is said to be bidentate. Thus the only ligand. which is expected to be bidentate is C_2OCl^{2-} as -



Hence, option (4) is correct.

90) IUPAC
$[Ag(H_2O)_2]$ $[Ag(CN)_2]$
Coordination number = 2,
Oxidation state = Ag^{+1}
Diaquasilver(I) dicyanidoargentate(I)

BIOLOGY

91) NCERT Pg. # 59

92) NCERT-XII, Pg. # 71

93)

NCERT, Pg. # 61

94)

NCERT, Pg. # 65

95)

NCERT, Pg. # 63

96)

NCERT, Pg. # 69

97)

NCERT-XII, Pg # 89, 90

98) NCERT Pg. # 75, 76

99) NCERT Pg. # 83

100) NCERT Pg. # 75(E), 83(H)

101)

NCERT, Pg. # 74

102) NCERT Page No. # 96

103)

NCERT, Pg. # 81

104) NCERT, Pg.No.101,102

105)

NCERT-XII, Pg # 107

106) NCERT, Pg # 94(E), 104(H)

107) NCERT Page No. # 117

108) NCERT Page No. # 111/112

109)

NCERT, Pg. # 95, 99

110)

NCERT, Pg. # 89

111) NCERT-XII, Pg. No. # 98, 99

112)

NCERT, Pg. # 93, 96, 98

113) NCERT-XII, Pg # 98

114) NCERT Pg # 108, Fig. 6.9

115)

NCERT, Pg. # 83

116)

NCERT Pg.# 236,237

117) NCERT XII Pg. # 199

118)

NCERT, Pg. # 193-194

119)

NCERT, Pg. # 190

120)

NCERT Pg # 192, Line 9th

121)

NCERT XII, Pg. No. 231-232

122)

NCERT (XII) Pg. # 230

123) NCERT XII Page No. : 197

124) NCERT XIIth Page No. 199

125)

NCERT-XII, Pg. # 245

126)

NCERT Page No. 212

127)

NCERT XII, Page # 214

128)

NCERT-XII, Pg. # 210

129)

NCERT, Pg. # 207

130)

NCERT, Pg. # 212

131)

NCERT-XII, Pg.# 225

132) NCERT (XII) Pg. # 261, 267, 284

133)

NCERT-XII, Pg. # 218

134)

NCERT, Pg. # 219

135)

NCERT, Pg. # 218-219

136)

NCERT, Pg. # 201

OR

Bacterial Artificial Chromosome (BAC) is a type of cloning vector, so its use is a vector-mediated approach.
Final Answer: A & B both

137)

NCERT, Pg. # 200

138)

NCERT, Pg. # 198

139)

NCERT, Pg. # 195, 201, 202

140)

NCERT-XII, Pg # 198

141) NCERT XII Pg # 198(E), 217(H)

142) NCERT XII Pg # 201 (E)

143)

NCERT, Pg. # 209

144)

NCERT, Pg. # 213

145) NCERT-XII, Pg. # 209

146)

NCERT, Pg. # 208, 209

147)

NCERT, Pg. # 209, 211, 213

148) NCERT (XIIth) Pg. # 208

149)

NCERT, Pg. # 211

150)

NCERT, Pg. # 187

151) NCERT Pg. # 181,183, 185

152)

NCERT Pg. # 187, 188

153)

NCERT, Pg. # 182

154)

NCERT XII Pg # 155

155) NCERT Pg#134

156)

NCERT Pg # 141

157) NCERT Pg#143

158) NCERT, Pg. # 143

159) NCERT, Pg. # 131(E)

160) NCERT Pg. # 131

161)

NCERT, Pg. # 135, 136

162) NCERT Pg#131,132

163) NCERT Pg # 157

164) NCERT Pg. # 137(E), 150(H)

165)

NCERT, Pg. # 136

166)

NCERT, Pg. # 145

167) NCERT Pg. # 135,136(E),147,149(H)

168)

NCERT, Pg. # 132

169) NCERT Page No. 160

170) NCERT-XII, Pg. # 117

171) NCERT page No. - 124-125

172)

Module

173) NCERT_Page No.140,141

174) NCERT pg no # 123

175) NCERT Pg. No. # 140-142

176) NCERT Pg. # 110, 111

177) NCERT Pg # 116

178)

NCERT Pg. # 116

179) NCERT Pg. # 113 (E) / 125 (H)

180)